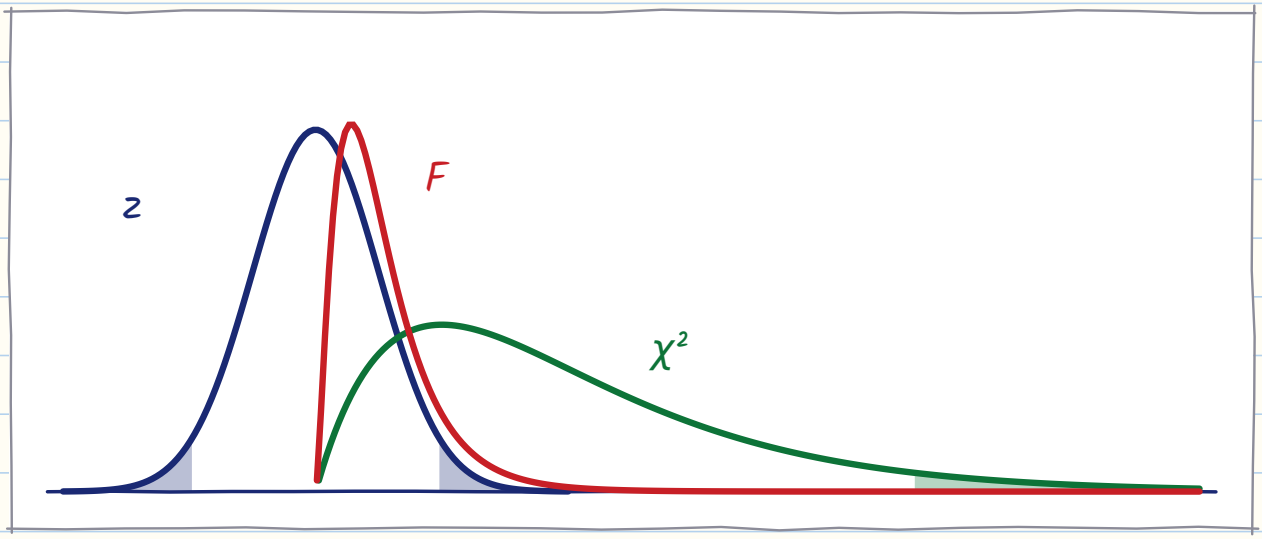
$z \cdot \chi^2 \cdot F$

Handwritten-Style Study Notes

z-Test · Chi-Square Test · One-Way & Two-Way ANOVA

math intuition · logic · Excel · Python



normal (z), chi-square and F: three curves, one idea

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statistics notebook, part 2

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How to read these notes

- **Blue ink** = main notes. **Red** = key terms and formulas.
- Green = tips. Purple boxes = intuition (the WHY). Orange = rules of thumb.
- Every test has: definition, formula, intuition, steps, worked numbers, Excel, Python.
- Section 19 = 10 practice problems. Section 20 = answer key.

MASTER IDEA

Master skeleton: every test compares what you OBSERVED with what H_0 EXPECTS, scaled by noise. Big ratio = surprising = evidence against H_0.

1. The Big Picture: z, χ^2 and F

three families, one idea: is the pattern bigger than chance?

The map

Family	Data type	Core question	Statistic	Curve
z-test	means (σ known) or proportions	is a mean/proportion off the claim, or are two groups different?	z	normal
Chi-square	COUNTS in categories	do counts match a theory? are two categories linked?	χ^2	χ^2
ANOVA	means of 3+ groups	are ALL the group means equal?	F	F

- **z**: gap between observed and claimed value, counted in standard errors.
- **χ^2**: total squared gap between observed counts and expected counts.
- **F**: variation BETWEEN groups divided by variation WITHIN groups.

How they connect to each other (and to t)

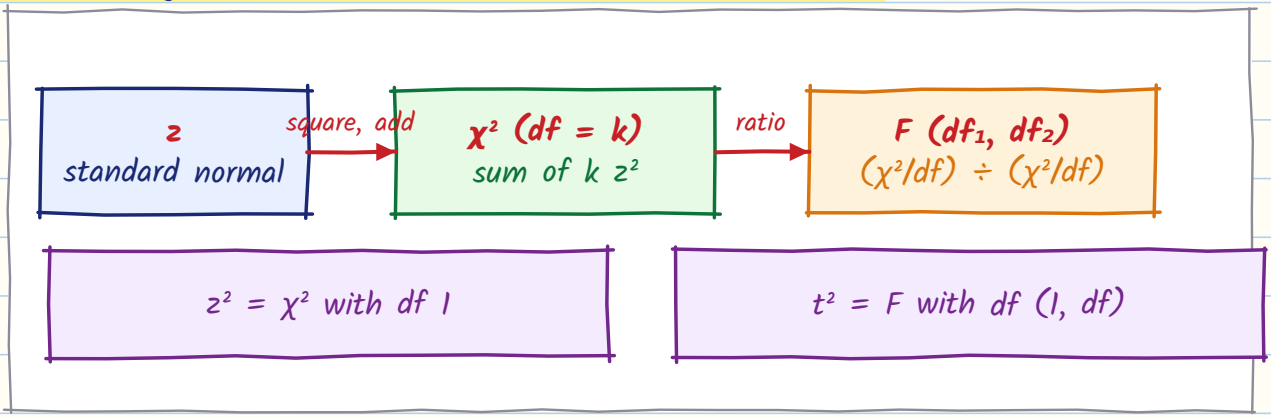


Fig 1: the family tree of test distributions.

- Square a z-score and you get χ^2 with 1 df. Add k of them → χ^2 with k df.
- Divide two χ^2 (each by its df) → F.
- t-test with 2 groups and one-way ANOVA agree: $F = t^2$.
- z is the t-test's big-sample limit: as df grows, t → z.

RULE

When σ is KNOWN use z. When σ is estimated from the sample use t (earlier notes). For 3+ means use ANOVA. For counts use χ^2. For success/fail rates use the proportion z.

The universal skeleton

Test	observed vs expected	noise scaling
z-test	$\bar{x} - \mu_0$ (or $\hat{p} - p_0$)	divide by the standard error
χ^2 test	$(O - E)$ in every cell	square it, divide by E, add up
ANOVA	spread of group means	divide by spread inside groups

> **INTUITION**
> Statistics is signal vs noise. All three tests output a number that is small when H_0 is true and large when something real is going on. We then ask how rare such a large number would be by luck alone.

Vocabulary refresher (quick)

- **Parameter** = true value in the population (μ, p, σ). **Statistic** = from the sample ($\bar{x}$, $\hat{p}$, s).
- **H_0** = nothing going on. **H_1** = something is. **α** = false-alarm rate (usually 0.05).
- **p-value** = chance of a result at least this extreme if H_0 were true.
- $p < \alpha \rightarrow$ Reject H_0. $p \geq \alpha \rightarrow$ Fail to reject H_0 (never 'accept').
- z is symmetric (two tails possible). χ^2 and F use only the RIGHT tail.

2. Foundations: Normal, χ^2 and F Curves

normal curve, z-scores, sampling distributions, χ^2 and F

The normal distribution

- Symmetric bell, fully described by its mean μ and SD σ.

within	$\mu \pm 1\sigma$	$\mu \pm 2\sigma$	$\mu \pm 3\sigma$	$\pm 1.96\sigma$	$\pm 2.576\sigma$
area	68.27%	95.45%	99.73%	95%	99%

z-score = distance in SDs

$$z = (x - \mu) / \sigma$$

standardise to mean 0, SD 1

- IQ scores: $\mu = 100$, $\sigma = 15$. A score of 130 has $z = (130 - 100)/15 = 2.00$.
- $P(IQ > 130) = P(z > 2) = 0.0228$. Only about 2.3% of people.
- $z = 0$ means average. $|z| > 2$ means unusual. $|z| > 3$ means very rare.

INTUITION

Standardising erases units. A z of 2 means the same thing for heights, marks or milliseconds: 'two SDs from typical'.

Critical z values to memorise

α	0.10	0.05	0.01
two-tailed $\pm z$	1.645	1.960	2.576
one-tailed z	1.282	1.645	2.326

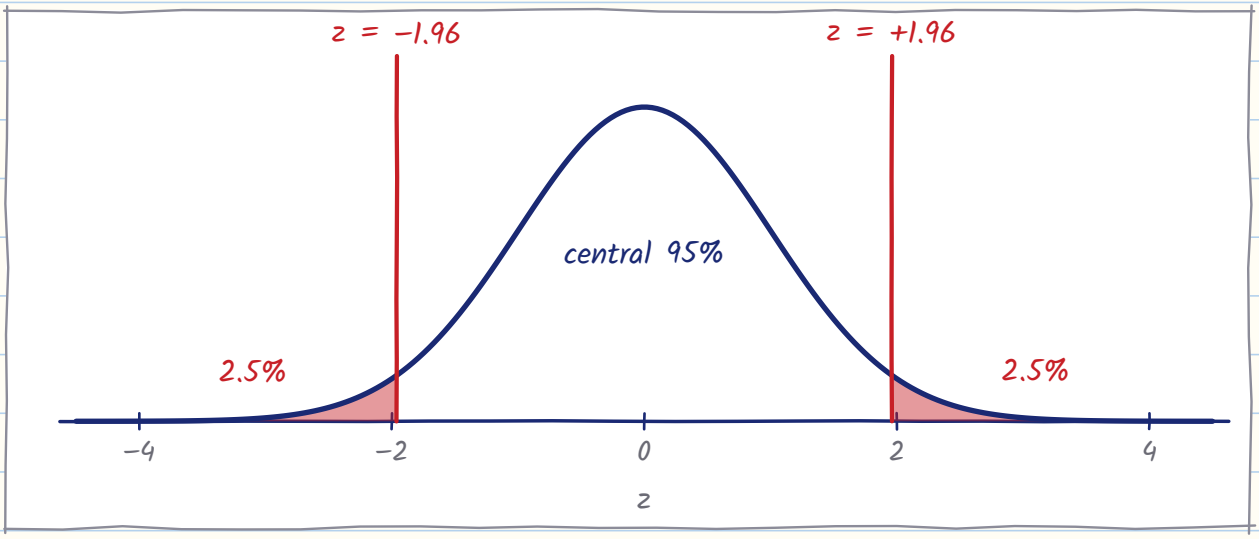


Fig 2: the 5% two-tailed rejection zone of the standard normal.

Sampling distribution of the mean

- Take many samples of size n. Their means form a distribution.
- Centre = μ. Spread = **standard error** σ/√n. Shape ≈ normal (CLT).

$$z = (\bar{x} - \mu) \,/\, (\sigma/\sqrt{n})$$

the z-test statistic

Sampling distribution of a proportion

- $\hat{p} = x/n$ = fraction of successes in the sample.
- Centre = p. Spread = $\sqrt{[\, p(1-p)/n \,]}$. Shape ≈ normal if $np \geq 10$ and $n(1-p) \geq 10$.

NOTE

For a proportion the SD is built from p itself, so no separate σ is needed. That is why proportion tests use z, never t.

The chi-square distribution

$$\chi^2_k = Z_1^2 + Z_2^2 + \ldots + Z_k^2$$

k squared standard normals

- Squaring removes the sign, so $\chi^2 \geq 0$ and every deviation ADDS to the total.
- Mean = df. Variance = 2·df. Right-skewed; skew fades as df grows.
- Only the RIGHT tail matters: a big total gap is the only sign of trouble.

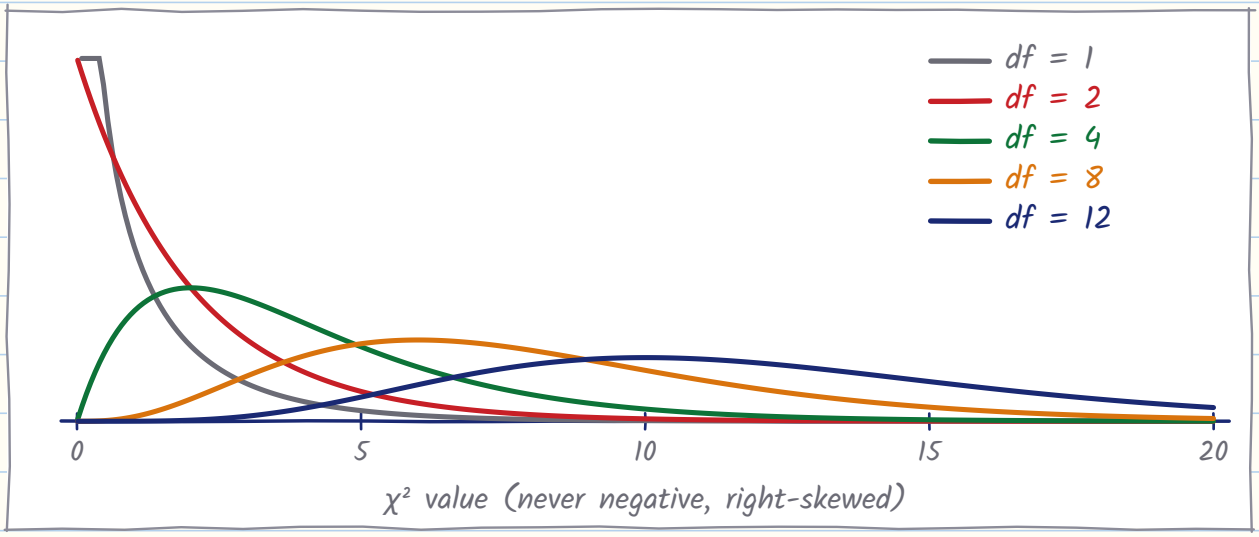


Fig 3: χ^2 curves for several df. Bigger df shifts right and flattens.

The F distribution

$$F = (\chi^2_1/df_1) \;/\; (\chi^2_2/df_2)$$

ratio of two variance estimates

- If H_0 is true, both numerator and denominator estimate the SAME σ^2. Ratio ≈ 1.
- $F \geq 0$, right-skewed, needs TWO df: numerator df_1 and denominator df_2.
- Only the right tail is used: F much bigger than 1 means the group means differ.

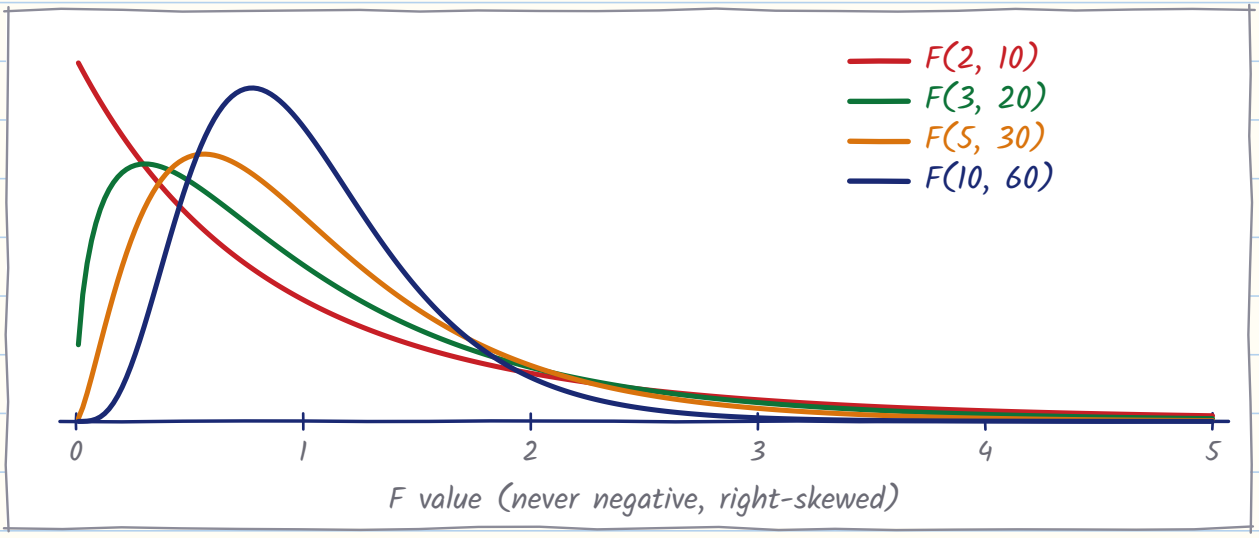


Fig 4: F curves. Peak sits near 1 when df_2 is large.

Critical-value lookups (Excel / Python)

Need	Excel	SciPy
z critical (two-tailed 5%)	NORM.S.INV(0.975)	stats.norm.ppf(0.975)
p from z (right tail)	1-NORM.S.DIST(z,TRUE)	stats.norm.sf(z)
χ^2 critical (5%)	CHISQ.INV.RT(0.05,df)	stats.chi2.ppf(0.95, df)
p from χ^2	CHISQ.DIST.RT(x,df)	stats.chi2.sf(x, df)
F critical (5%)	F.INV.RT(0.05,df1,df2)	stats.f.ppf(0.95, d1, d2)
p from F	F.DIST.RT(F,df1,df2)	stats.f.sf(F, d1, d2)

3. CORE DEFINITIONS: The Eight Tests

definition, H_0, formula and df for each of the eight tests

> **PLAIN ENGLISH**
> Plain English. z-tests: 'is this mean or proportion different?' χ^2 tests: 'do the COUNTS fit?' ANOVA: 'do ALL the group means agree?'

1) One-sample z-test (mean, σ known)

- **Definition**: tests one population mean against a claim μ_0 when σ is known.
- H_0: $\mu = \mu_0$. Curve: standard normal.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

2) Two-sample z-test (two means, σ_1 and σ_2 known)

- **Definition**: compares the means of two independent groups with known SDs.
- H_0: $\mu_1 = \mu_2$.

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\sigma_1^2/n_1 + \sigma_2^2/n_2 \right)}}$$

3) One-proportion z-test

- **Definition**: tests whether a population proportion equals a claim p_0.
- H_0: $p = p_0$. Needs $np_0 \geq 10$ and $n(1-p_0) \geq 10$.

$$z = \frac{\hat{p} - p_0}{\sqrt{\left[p_0(1-p_0) / n \right]}}$$

4) Two-proportion z-test

- **Definition**: compares the success rates of two independent groups.
- H_0: $p_1 = p_2$. Uses the POOLED proportion $\hat{p} = (x_1+x_2)/(n_1+n_2)$.

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{[\ \hat{p}(1-\hat{p})(1/n_1 + 1/n_2)\ ]}}$$

5) Chi-square goodness-of-fit

- **Definition**: do observed counts in k categories match expected proportions?
- H_0: the distribution is as claimed. df = k − 1 (minus parameters estimated).

$$\chi^2 = \frac{(O - E)^2}{E} \text{ summed over all categories}$$

6) Chi-square test of independence

- **Definition**: are two categorical variables related? Uses a contingency table.
- H_0: the variables are independent. E = (row total × column total)/N.

$$\chi^2 = \Sigma\Sigma\ (O - E)^2 / E\ ,\ df = (r - 1)(c - 1)$$

7) One-way ANOVA

- **Definition**: tests whether k ≥ 3 group means are all equal (ONE factor).
- H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$. H_1: at least one mean differs.

$$F = \frac{MS_{between} = SSB/(k-1)}{MS_{within} = SSW/(N-k)}\ ,\ df = (k-1,\ N-k)$$

8) Two-way ANOVA

- **Definition**: tests the effects of TWO factors and their interaction on a mean.
- Three tests: main effect A, main effect B, interaction A×B. Each F = MS_effect / MS_within.

$$F_A = MS_A/MS_W \;,\; F_B = MS_B/MS_W \;,\; F_{AB} = MS_{AB}/MS_W$$

Side-by-side summary

Test	Data	Statistic	df
One-sample z	one mean, σ known	z	(none: normal)
Two-sample z	two means, σs known	z	(none: normal)
One-prop z	one proportion	z	(none: normal)
Two-prop z	two proportions	z	(none: normal)
χ^2 GOF	counts, 1 variable	χ^2	$k - 1$
χ^2 independence	counts, 2 variables	χ^2	$(r-1)(c-1)$
One-way ANOVA	k means, 1 factor	F	$(k-1, N-k)$
Two-way ANOVA	means, 2 factors	F (x3)	see Section 8

TIP

Hidden links: χ^2 independence on a 2×2 table equals the squared two-proportion z ($z^2 = \chi^2$). One-way ANOVA on 2 groups equals the pooled t-test squared ($F = t^2$).

4. Z-Tests: Theory and Intuition

means and proportions, standard errors, and the pooling trick

When is z allowed?

- **σ is genuinely known**: calibrated machine SD, a long-run process SD, standardised tests such as IQ ($\sigma = 15$).
- **Proportions**: the SD comes from p itself, so z is the natural tool.
- Large n (say > 30) lets z approximate t, but t is still more correct if σ is estimated.

> **WATCH OUT**
>
> Beginner trap: textbook says 'n ≥ 30 use z'. Real practice: if you only have s, use t. z needs a KNOWN σ.

One-sample z-test: the logic

- Claim: the true mean is μ_0. Under H_0 the sample mean is normal around μ_0 with $SE = \sigma/\sqrt{n}$.
- z = how many standard errors the observed $\bar{x}$ sits from μ_0.

1. H_0: $\mu = \mu_0$. H_1: $\mu \neq$, >, or < μ_0.
2. $SE = \sigma/\sqrt{n}$. $z = (\bar{x} - \mu_0)/SE$.
3. p-value from the normal curve (tails below). Compare with α.
4. State the conclusion in plain words.

H_1	p-value formula	reject if
$\mu \neq \mu_0$ (two-tailed)	$2 \times P(Z > \lvert z \rvert)$	$\lvert z \rvert > z_{\alpha/2}$
$\mu > \mu_0$ (right-tailed)	$P(Z > z)$	$z > z_\alpha$
$\mu < \mu_0$ (left-tailed)	$P(Z < z)$	$z < -z_\alpha$

Confidence interval for μ (σ known)

$$\bar{x} \pm z_{\alpha/2} \times \sigma/\sqrt{n}$$

95%: $z = 1.96$

Two-sample z-test: where the SE comes from

- Each sample mean has variance σ^2/n. The groups are independent.
- Variances ADD even when you subtract the means:

$$Var(\bar{x}_1 - \bar{x}_2) = \sigma_1^2/n_1 + \sigma_2^2/n_2$$

INTUITION

Subtracting two noisy numbers gives a noisier number. Uncertainty from both groups accumulates, never cancels.

Proportion z-tests: the one detail that matters

- **One-proportion**: under H_0 the true p is p_0, so the SE uses p_0: $\sqrt{[p_0(1-p_0)/n]}$.
- **Two-proportion**: H_0 says $p_1 = p_2$, i.e. ONE common rate. Best estimate = pooled $\hat{p}$.

$$\hat{p} = (x_1 + x_2) / (n_1 + n_2)$$

pool the successes

- For a CONFIDENCE INTERVAL we do not assume H_0, so use each group's own $\hat{p}(1-\hat{p})/n$ (unpooled).

RULE

Memory hook: TEST assumes H_0 is true → pool. INTERVAL makes no such assumption → do not pool.

Conditions (check them!)

- Random sample, independent observations.
- Means: σ known and data normal, or n large. Proportions: successes and failures both ≥ 10.

Picture of a z-test decision

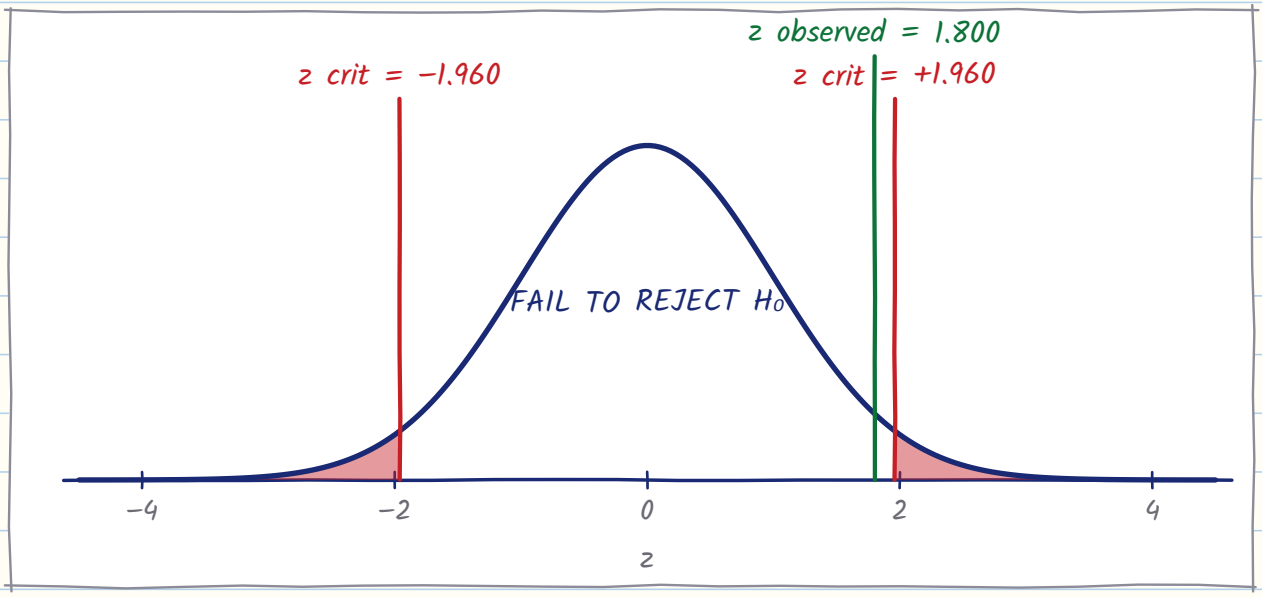


Fig 5: z = 1.80 lies inside the central 95% (two-tailed), so fail to reject.

How big a sample do I need?

$$n = (z_{\alpha/2}\ \sigma / E)^2 \text{ (mean)}$$

$$n = z_{\alpha/2}^2\ p(1-p) / E^2 \text{ (proportion)}$$

- E = tolerable margin of error. Use p = 0.5 when unknown (worst case).
- Poll with E = 3 points, 95%: $n = 1.96^2 \times 0.25 / 0.03^2 = 1068$ people.
- Halving E quadruples n. Precision is expensive.

5. Chi-Square Tests: Theory and Intuition

observed counts vs expected counts

The idea

- You have COUNTS in categories (not measurements). Theory predicts what the counts should be.
- **O** = observed count. **E** = expected count if H_0 were true.

$$\chi^2 = \Sigma\ (O - E)^2\ /\ E$$

Why that exact formula?

- **O − E**: the raw gap in each cell.
- **Square it**: plus and minus gaps would cancel to 0. Squaring stops that.
- **Divide by E**: a gap of 5 matters a lot when E = 10, almost nothing when E = 1000.
- **Add up**: one total 'misfit score'. Zero = perfect fit. Bigger = worse fit.

INTUITION

Link to the normal curve. For a count with large E, $(O - E)/\sqrt{E}$ is roughly standard normal. Square and add these → χ^2 distribution. That is exactly why the statistic follows a chi-square curve.

Degrees of freedom

- GOF: df = k − 1. Counts must add to N, so only k − 1 are free.
- If you estimated m parameters from the data to get E (say a Poisson mean): df = k − 1 − m.
- Independence: df = (r − 1)(c − 1). Once row and column totals are fixed, only that many cells are free.

Goodness of fit: steps

1. H_0: the distribution is as claimed (e.g. all faces equally likely, or 9:3:3:1).
2. $E_i = N \times$ claimed proportion$_i$. Check every $E_i \geq 5$.
3. $\chi^2 = \Sigma (O-E)^2/E$, $df = k - 1$.
4. Compare with χ^2 critical (right tail), or $p = P(\chi^2_{df} > \chi^2)$.

Independence: where E comes from

- If the variables are independent, P(row i and column j) = P(row i) × P(col j).

$$E_{ij} = (\text{row total}_i \times \text{column total}_j) \,/\, N$$

INTUITION

Independence in words: knowing the row tells you nothing about the column. Then each row has the SAME column mix as the overall table. E builds exactly that table.

Independence vs homogeneity

- Independence: ONE sample, classify each person by two variables.
- Homogeneity: SEVERAL samples fixed in advance, compare their category mix.
- Same arithmetic, same χ^2, different wording of H_0.

Conditions

- Counts, not percentages. Observations independent. Each person counted once.
- Expected counts: all $E \geq 5$ (relaxed: at least 80% of cells ≥ 5 and none < 1).
- Small E? Merge categories, collect more data, or use Fisher's exact test (2×2).

WATCH OUT

Paired / repeated yes-no data (same people before and after) is NOT independent. Use McNemar's test, not χ^2.

2×2 tables and Yates correction

$$\chi^2_{Yates} = \Sigma \left(|O - E| - 0.5 \right)^2 / E$$

- Shrinks each gap by 0.5 to compensate for discrete counts. Conservative.
- Only for 2×2 (df = 1). Many modern texts skip it when E is comfortably large.

Effect size and finding the culprit cell

- **Cramér's V** = √[χ² / (N × min(r−1, c−1))]. 0 = no link, 1 = perfect.
- For 2×2, V = φ (phi). Rough sizes: 0.1 small, 0.3 medium, 0.5 large (df* = 1).
- **Standardised residual** = (O − E)/√E. Cells with |value| > 2 drive the result.

> **TIP**
>
> Big N makes tiny links 'significant'. Always report V next to χ².

Picture of a χ² decision

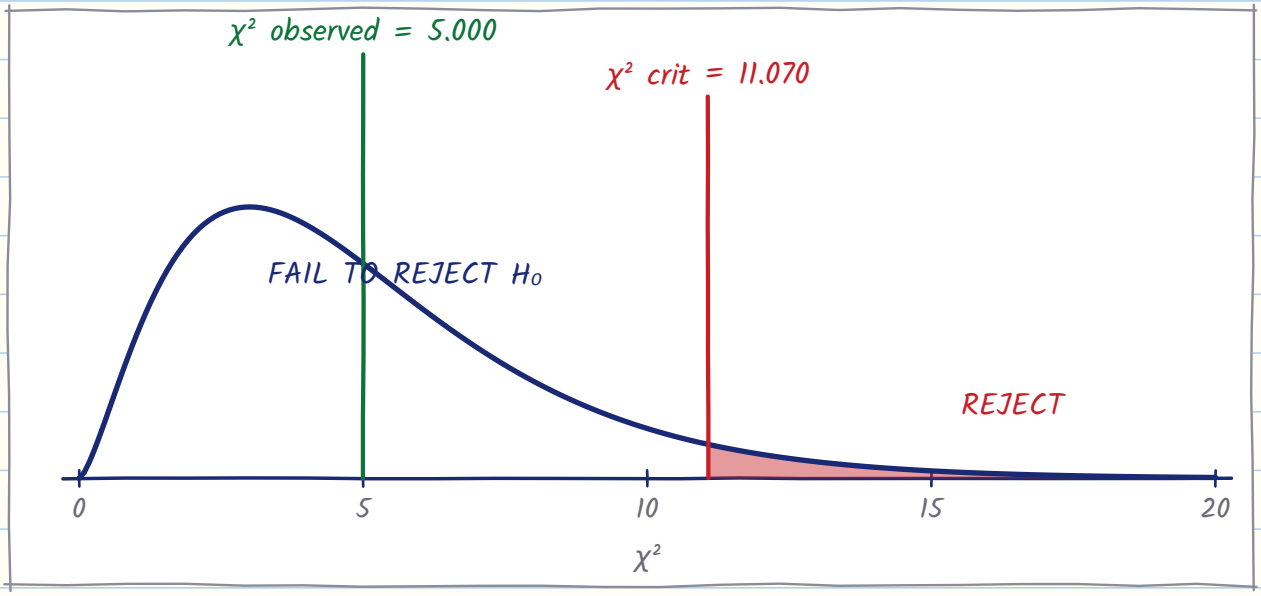


Fig 6: df = 5. Observed χ² = 5.0 is far from the 11.07 cutoff.

Bonus: χ² test for ONE variance

$$\chi^2 = (n - 1)\, s^2 / \sigma_0^2 \,, \quad df = n - 1$$

- Tests whether a population variance equals a claimed σ_0^2. Needs normal

data (fragile).

6. ANOVA Theory: Splitting the Variance

why ANOVA, between vs within, and the F-ratio

The problem with many t-tests

- Comparing 3 groups by t-tests means 3 comparisons (A–B, A–C, B–C).
- Each test risks 5% false alarm, and the risks pile up:

groups k	2	3	4	5	6
pairwise tests m	1	3	6	10	15
P(at least 1 false alarm)	5.0%	14.3%	26.5%	40.1%	53.7%

WATCH OUT

Familywise error = 1 − (1 − α)^m. ANOVA asks ONE question first ('are all means equal?') with ONE test at α = 5%.

The big idea: compare two kinds of spread

- **Between-group variation**: how far the group means are from each other.
- **Within-group variation**: how much individuals vary inside their own group (pure noise).
- If between is much bigger than within, the groups really differ.

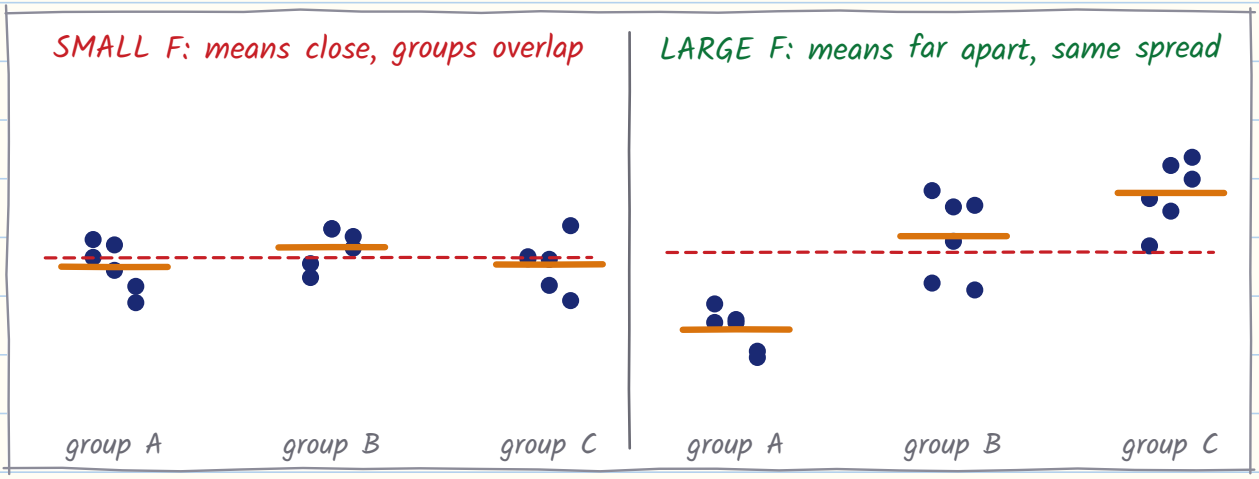


Fig 7: same noise inside groups; only the gaps between means change.

Hypotheses

- H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$ (all group means equal).
- H_1: at least ONE mean is different. (Not 'all are different'!)

Partitioning the total variation

$$SST = SSB + SSW$$

total = between + within

- **SST** = $\Sigma\Sigma$ $(x - \text{grand mean})^2$ total spread of all N observations.
- **SSB** = $\Sigma\, n_j\, (\bar{x}_j - \text{grand mean})^2$ spread of group means (signal).
- **SSW** = $\Sigma\Sigma$ $(x - \bar{x}_j)^2$ spread inside groups (noise).
- df partition: $N - 1 = (k - 1) + (N - k)$.

Mean squares and F

$$MSB = SSB/(k-1)\ ,\ MSW = SSW/(N-k)\ ,\ F = MSB\ /\ MSW$$

INTUITION

Why $F \approx 1$ under H_0. MSW ALWAYS estimates the noise variance σ^2. MSB estimates σ^2 only when all means are equal; if not, MSB = σ^2 + (extra from real mean gaps). So $F = (\sigma^2 + \text{extra})/\sigma^2$ is larger than 1 only when real differences exist.

The ANOVA table (template)

Source	SS	df	MS	F
Between	SSB	$k - 1$	SSB/(k−1)	MSB/MSW
Within	SSW	$N - k$	SSW/(N−k)	
Total	SST	$N - 1$		

Steps

1. H_0: all means equal. Choose α.
2. Compute group means and the grand mean.
3. SSB, SSW (and SST as a check).

4. MSB, MSW, then F = MSB/MSW with df (k−1, N−k).
5. Compare with F critical (right tail) or p. If significant → post-hoc (next section).

Assumptions

- **Independence** of observations.
- **Normality** of residuals (values minus their group mean), not of the pooled data.
- **Equal variances** (homogeneity). Rule: largest SD / smallest SD < 2, or Levene $p > 0.05$.
- ANOVA is quite robust to mild violations when group sizes are equal.

> **TIP**
> With only 2 groups ANOVA gives $F = t^2$ (pooled t). Same p-value, same decision.

7. After ANOVA: Post-Hoc, Effect Size, Assumptions

which groups differ? how big? what if assumptions fail?

A significant F only says 'somewhere'

- Reject H_0 means at least one mean differs. It does NOT say which.
- Follow up with a **post-hoc test** that controls the familywise error.

Tukey's HSD (Honestly Significant Difference)

- Compares ALL pairs while keeping the overall error at α. Best for equal n.

$$HSD = q_{crit} \times \sqrt{(MSW / n)}$$

n = size of each group

- q_crit comes from the studentised-range table for (k, df_within).
- Any pair whose mean gap exceeds HSD is significantly different.

> **NOTE**
>
> Equivalent: Tukey $q = |\bar{x}_i - \bar{x}_j| / \sqrt{(MSW/n)}$. Compare q with q_crit.

Other post-hoc / follow-up options

Method	Use when
Tukey HSD	all pairwise comparisons, equal n
Tukey–Kramer	all pairs, unequal n
Bonferroni	a few planned comparisons: use α/m
Games–Howell	unequal variances (pairs with Welch ANOVA)
Dunnett	every group vs ONE control group
Planned contrasts	specific hypotheses stated before data

Effect size for ANOVA

$$\eta^2 = SSB / SST$$

share of total variation explained

- $\eta^2 = 0.66$ means 66% of the variation in the outcome is explained by the groups.

- Benchmarks: 0.01 small, 0.06 medium, 0.14 large.

$$\omega^2 = (SSB - df_B \times MSW) / (SST + MSW)$$ less biased

- ω^2 shrinks η^2 slightly to correct small-sample optimism. Report either, say which.
- Partial η^2 (two-way) = SS_effect / (SS_effect + SSW).

Checking the assumptions

Assumption	Check	Rule
Normal residuals	Q-Q plot, Shapiro–Wilk	$p > 0.05$ ok
Equal variances	Levene / Bartlett	$p > 0.05$ ok
Independence	study design	no repeated subjects
No wild outliers	box plots	inspect per group

When assumptions fail

Problem	Remedy
Unequal variances	Welch ANOVA + Games–Howell
Non-normal, small n	Kruskal–Wallis test (ranks)
Same subjects in every group	repeated-measures ANOVA
Wildly skewed data	log / sqrt transform first
Many outcomes at once	MANOVA

How to report

REPORT FORMAT

$F(2, 15) = 14.77$, $p < .001$, $\eta^2 = 0.66$. Tukey HSD: B exceeded A and C; A and C did not differ. Always give df, F, p, effect size, and the post-hoc result.

8. Two-Way ANOVA Theory

two factors, three questions: A, B and A×B

The setup

- Two factors: **A** with a levels, **B** with b levels.
- Every combination is a **cell**. Replication n = observations per cell. Total N = a·b·n.
- Example: teaching style (Interactive, Lecture) × study time (Low, Medium, High).

Three questions, three F-tests

1. **Main effect of A**: ignoring B, do the A-level averages differ?
2. **Main effect of B**: ignoring A, do the B-level averages differ?
3. **Interaction A×B**: does the effect of A CHANGE depending on the level of B?

The model

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

- μ = grand mean. α_i = effect of A-level i. β_j = effect of B-level j.
- $(\alpha\beta)_{ij}$ = extra effect when A-level i and B-level j occur TOGETHER.
- ε = random noise, assumed normal with equal variance in every cell.

Interaction: the key new idea

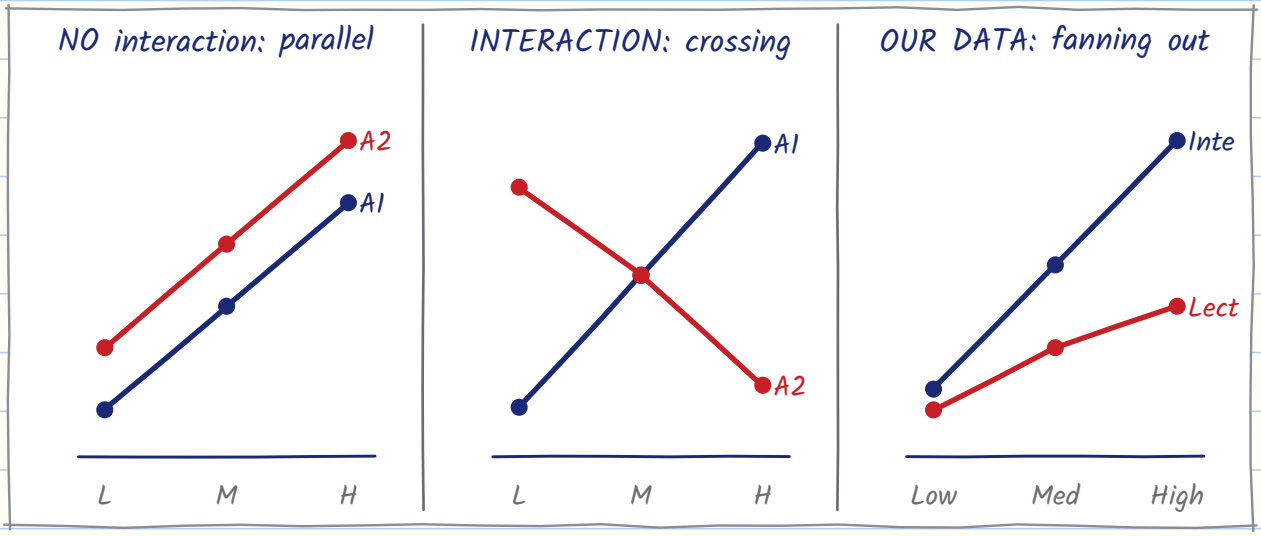


Fig 8: parallel lines = no interaction. Lines that cross or fan out = interaction.

- Parallel lines: A adds the same amount at every level of B. Effects simply ADD.
- Non-parallel lines: the benefit of A depends on B. 'It depends' = interaction.

INTUITION

Example. Interactive teaching helps a lot when students study HIGH hours, but barely matters when they study LOW hours. The gap between teaching styles grows with study time. That is an interaction.

Splitting the variation

$$SST = SSA + SSB + SSAB + SSW$$

Source	SS formula	df	MS	F
A	$b \cdot n \ \Sigma(\bar{x}_{i..} - M)^2$	$a - 1$	SSA/df	MSA/MSW
B	$a \cdot n \ \Sigma(\bar{x}_{j.} - M)^2$	$b - 1$	SSB/df	MSB/MSW
A×B	$n \ \Sigma(\bar{x}_{ij} - \bar{x}_{i..} - \bar{x}_{j.} + M)^2$	$(a-1)(b-1)$	SSAB/df	MSAB/MSW
Within	$\Sigma(x - \bar{x}_{ij})^2$	$ab(n - 1)$	SSW/df	
Total	$\Sigma(x - M)^2$	$abn - 1$		

- M = grand mean. $\bar{x}_{i..}$ = mean of A-level i. $\bar{x}_{j.}$ = mean of B-level j. $\bar{x}_{ij}$ = cell mean.
- All three F-tests share the SAME denominator MSW (pure within-cell

noise).

How the interaction term is built

- Predict each cell mean from the two main effects alone: $\bar{x}_{i..} + \bar{x}_{.j} - M$.
- The leftover (actual cell mean − that prediction) is the interaction.
- No interaction → leftovers ≈ 0 → SSAB ≈ 0.

Reading the results: order matters

1. Look at the interaction F FIRST.
2. No interaction: interpret main effects directly.
3. Significant interaction: main effects can mislead. Analyse SIMPLE effects (A within each B level).

> **WATCH OUT**
> A significant interaction can hide or reverse a main effect. Two lines crossing in an X can have identical marginal means yet strong effects.

Design variants

- **With replication** ($n \geq 2$ per cell): interaction can be tested. This is our main design.
- **Without replication** ($n = 1$): no within-cell noise exists; interaction and error are mixed. Assume no interaction and use it as error (randomised block design).
- **Unbalanced** (unequal n): sums of squares depend on order. Use Type II or III SS.

Assumptions and effect sizes

- Independence, normal residuals, equal variance across cells (Levene).
- Partial η^2 for each effect = SS_effect / (SS_effect + SSW).

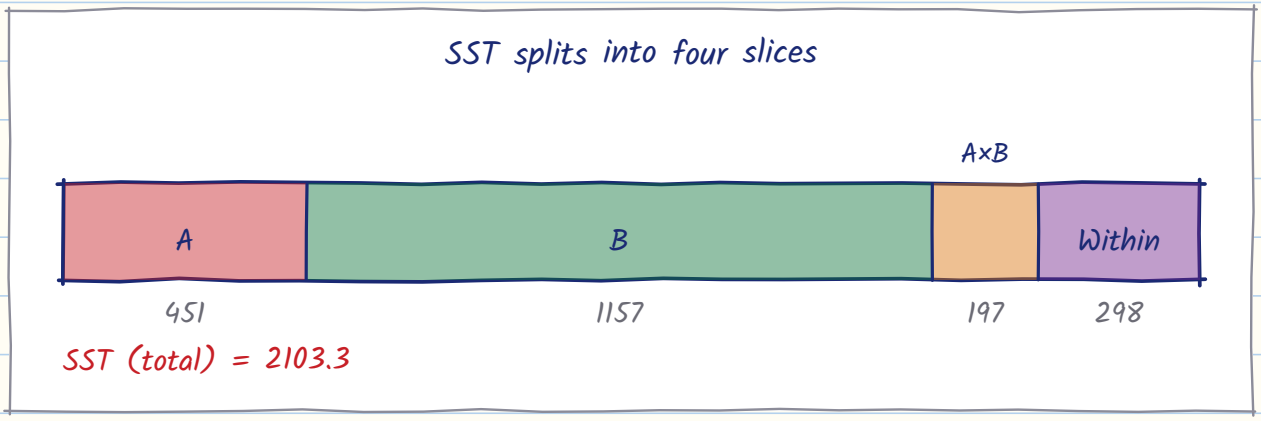


Fig 9: sums of squares for the teaching example (Section 13).

9. DIAGNOSTIC FLOW: Which Test?

quick, punchy rules: pick the right test in ten seconds

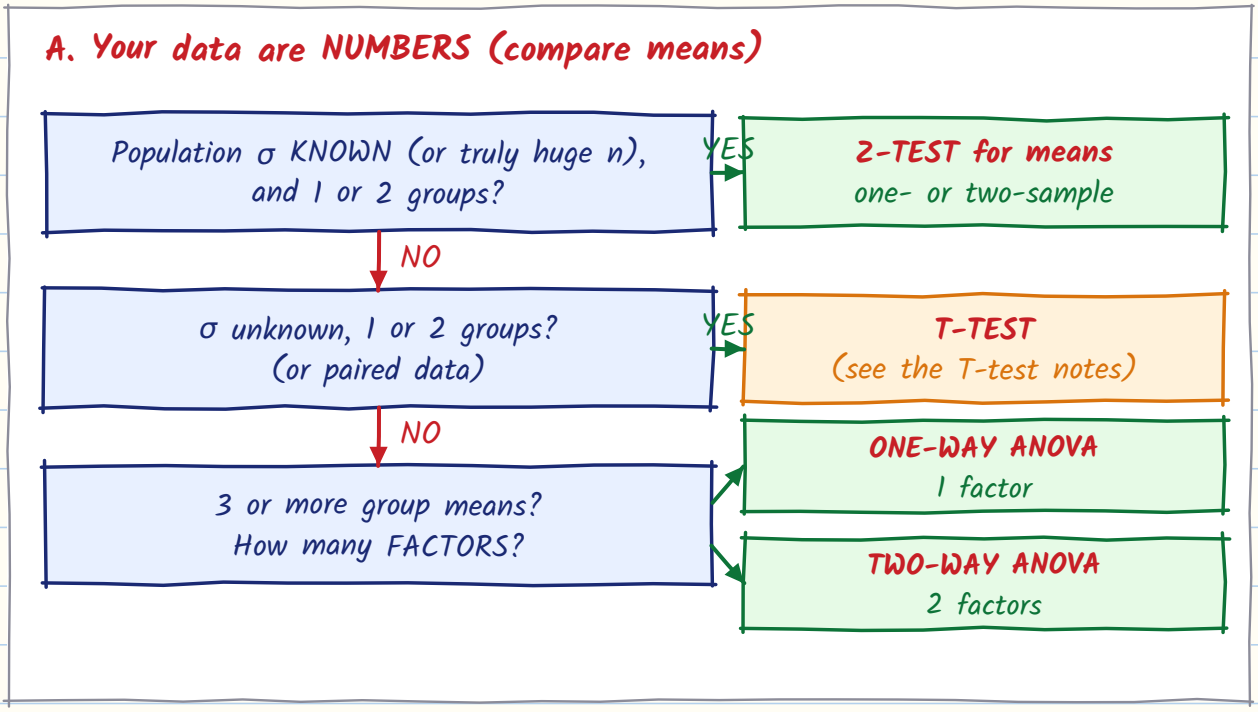


Fig 10: decision tree for NUMERICAL outcomes (means).

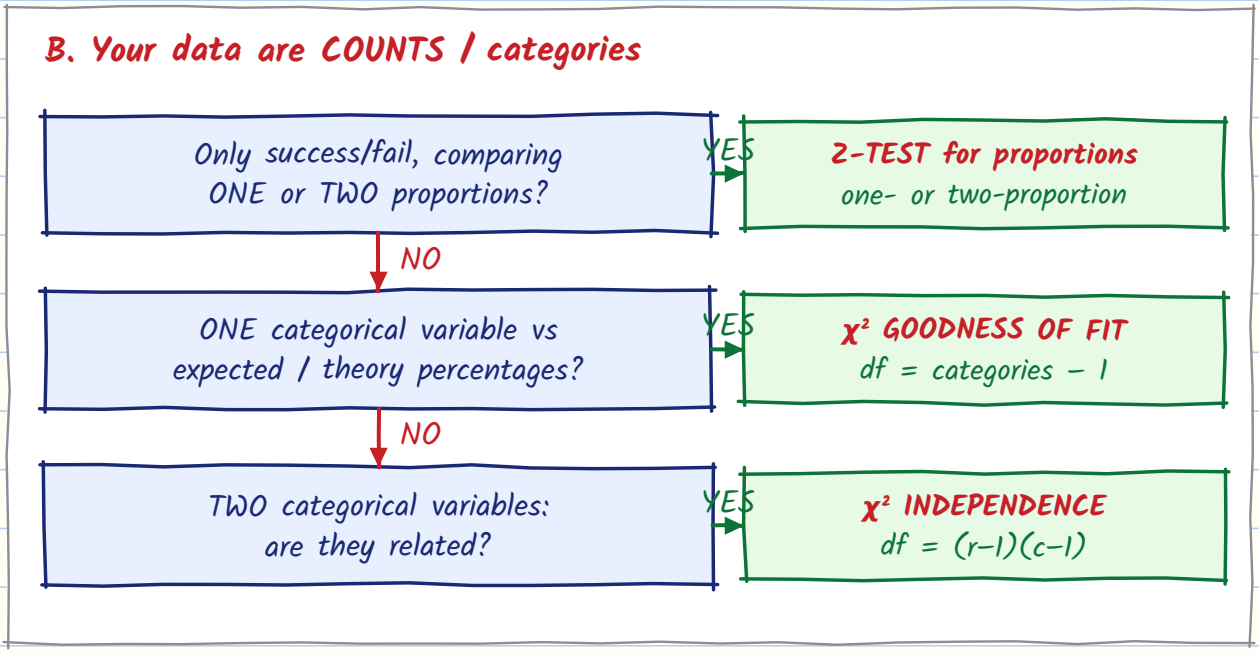


Fig 11: decision tree for COUNT / category outcomes.

Punchy rules

- Numbers + σ KNOWN + one or two groups? **z-test**.
- Numbers + σ unknown + one or two groups? **t-test** (earlier notes).
- Numbers + THREE or more groups? **ANOVA**. One factor = one-way. Two factors = two-way.
- Success/fail rate(s) of one or two groups? **proportion z**.
- Counts in several categories vs a theory? **χ^2 goodness of fit**.
- Counts cross-classified by two variables? **χ^2 independence**.
- Same people measured repeatedly? **paired t / repeated-measures ANOVA / McNemar**.

> **RULE**
>
> First question always: 'Is my outcome a measurement (mean) or a count (category)?' That single answer halves the menu.

Scenario drill

Scenario	Test
Class mean IQ vs 100, σ = 15 known	one-sample z
Two calibrated machines (σs known): mean fill	two-sample z
Did conversion beat the 10% target?	one-proportion z
Click rate of ad A vs ad B	two-proportion z
Is this die fair?	χ^2 goodness of fit
Do offspring follow 9:3:3:1?	χ^2 goodness of fit
Is smoking status linked to disease?	χ^2 independence
Yield of 3 fertilisers	one-way ANOVA
Teaching style × study time on marks	two-way ANOVA
Two means, σ unknown	t-test
Same patients, yes/no before and after	McNemar
3 skewed groups, tiny n	Kruskal–Wallis

10. NUMERICAL WORKING A: Z-Tests

four full scenarios: one mean, two means, one and two proportions

A1 One-sample z: class IQ

IQ scores have $\sigma = 15$ (known). A class of $n = 36$ has mean 104.5. Is the class mean different from 100? $\alpha = 0.05$.

1. H_0: $\mu = 100$. H_1: $\mu \neq 100$ (two-tailed).
2. $SE = 15/\sqrt{36} = 15/6 = \mathbf{2.50}$
3. $z = (104.5 - 100)/2.5 = 4.5/2.5 = \mathbf{1.80}$
4. Critical $z = \pm 1.960$. $|1.80| < 1.96$. p (two-tailed) $= 2 \times 0.0359 = \mathbf{0.0719}$
5. **FAIL TO REJECT H_0** at 5%.
6. 95% CI for $\mu = 104.5 \pm 1.96 \times 2.5 = (99.6, 109.4)$. It contains 100.

> **TAILS MATTER**
>
> What if the question was 'is the mean GREATER than 100?' (right-tailed). Then $p = 0.0359 < 0.05 \rightarrow$ REJECT. Same data, different decision: the tail choice must be made BEFORE seeing data.

```
B1  =(104.5-100)/(15/SQRT(36))                -> 1.80   (z)
B2  =2*(1-NORM.S.DIST(ABS(B1),TRUE))          -> 0.0719  (two-tailed p)
B3  =1-NORM.S.DIST(B1,TRUE)                   -> 0.0359  (right-tailed p)
B4  =NORM.S.INV(0.975)                        -> 1.9600 (z crit)
B5  =104.5-1.96*15/SQRT(36)  and  +           -> CI ends
Raw data version: =Z.TEST(A2:A37,100,15)      -> RIGHT-tail p only
```

A2 Two-sample z: two production lines

Line 1 ($\sigma_1 = 4$ known): $n = 40$, mean 52.3. Line 2 ($\sigma_2 = 5$ known): $n = 50$, mean 50.1. Do the means differ? $\alpha = 0.05$.

1. H_0: $\mu_1 = \mu_2$. H_1: $\mu_1 \neq \mu_2$.
2. $SE = \sqrt{(16/40 + 25/50)} = \sqrt{(0.4 + 0.5)} = \sqrt{0.9} = \mathbf{0.9487}$
3. $z = (52.3 - 50.1)/0.9487 = \mathbf{2.319}$
4. $p = 0.0204 < 0.05$, $|z| > 1.96 \rightarrow$ **REJECT H_0**.
5. 95% CI for $\mu_1 - \mu_2 = 2.2 \pm 1.96 \times 0.9487 = (0.34, 4.06)$.

```
=(52.3-50.1)/SQRT(4^2/40+5^2/50)              -> 2.319
=2*(1-NORM.S.DIST(ABS(z),TRUE))               -> 0.0204
ToolPak: 'z-Test: Two Sample for Means' (asks for the KNOWN variances)
```

A3 Two-proportion z: web page A/B test

Page A: 90 of 600 visitors bought (15.0%). Page B: 140 of 700 bought (20.0%). Is the purchase rate different? $\alpha = 0.05$, two-tailed.

1. $H_0: p_1 = p_2$. $H_1: p_1 \neq p_2$. Check counts: all successes and failures ≥ 10 ✓.
2. Pooled $\hat{p} = (90 + 140)/(600 + 700) = 230/1300 =$ **0.1769**
3. $SE = \sqrt{[\ 0.1769 \times 0.8231 \times (1/600 + 1/700)\ ]} =$ **0.0212**
4. $z = (0.150 - 0.200)/0.0212 =$ **−2.355**
5. $p = 0.0185 < 0.05 \rightarrow$ **REJECT H_0**. Page B converts better.
6. 95% CI (UNPOOLED SE = 0.0210): $-0.050 \pm 1.96 \times 0.0210 = (-0.091, -0.009)$.

> **NOTE**
> Pooled SE in the test (assumes H_0), unpooled SE in the interval (no assumption). Check: $z^2 = 5.547$ equals the χ^2 of the 2×2 table.

A4 One-proportion z: opinion poll

A mayor claims 50% support. In a poll of 400, 224 (56%) support. Is support different from 50%?

1. $H_0: p = 0.5$. Check: $np_0 = 200$, $n(1-p_0) = 200$, both ≥ 10 ✓.
2. $SE = \sqrt{[0.5 \times 0.5/400]} = \sqrt{0.000625} =$ **0.0250** (uses p_0, not $\hat{p}$)
3. $z = (0.56 - 0.50)/0.025 =$ **2.40** $p = 0.0164 \rightarrow$ **REJECT H_0**.

Python (all four)

```python
import numpy as np
from scipy import stats
from statsmodels.stats.proportion import proportions_ztest

def p2(z): return 2 * stats.norm.sf(abs(z))

z1 = (104.5 - 100) / (15 / np.sqrt(36))
```

```
z2 = (52.3 - 50.1) / np.sqrt(4**2 / 40 + 5**2 / 50)
print('A1 z =', round(z1, 3), ' p =', round(p2(z1), 4))
print('A2 z =', round(z2, 3), ' p =', round(p2(z2), 4))

# A3 two proportions (pooled, as in the test)
z3, p3 = proportions_ztest([90, 140], [600, 700])
print('A3 z =', round(z3, 3), ' p =', round(p3, 4))

# A4 one proportion, SE built from p0
z4, p4 = proportions_ztest(224, 400, value=0.5, prop_var=0.5)
print('A4 z =', round(z4, 3), ' p =', round(p4, 4))
# output:
# A1 z = 1.8  p = 0.0719
# A2 z = 2.319  p = 0.0204
# A3 z = -2.355  p = 0.0185
# A4 z = 2.4  p = 0.0164
```

II. NUMERICAL WORKING B: Chi-Square

goodness of fit (die, Mendel) and independence (study mode vs result)

B1 Goodness of fit: is the die fair?

A die is rolled 120 times. H_0: all six faces equally likely, so $E = 120/6 = 20$ each.

face	O	E	O − E	$(O-E)^2/E$
1	25	20	5	1.25
2	17	20	−3	0.45
3	15	20	−5	1.25
4	23	20	3	0.45
5	24	20	4	0.80
6	16	20	−4	0.80
sum	120	120	0	$\chi^2 = 5.00$

1. $df = k - 1 = 6 - 1 = 5$. χ^2 critical (5%) = **11.070**.
2. $\chi^2 = 5.00 < 11.070$ and $p = 0.4159$ → **FAIL TO REJECT H_0**.

> **NOTE**
> The die looks fair. Face counts wobble (15 to 25) but that is typical randomness for 120 rolls. Still no proof of fairness: only 'no evidence of bias'.

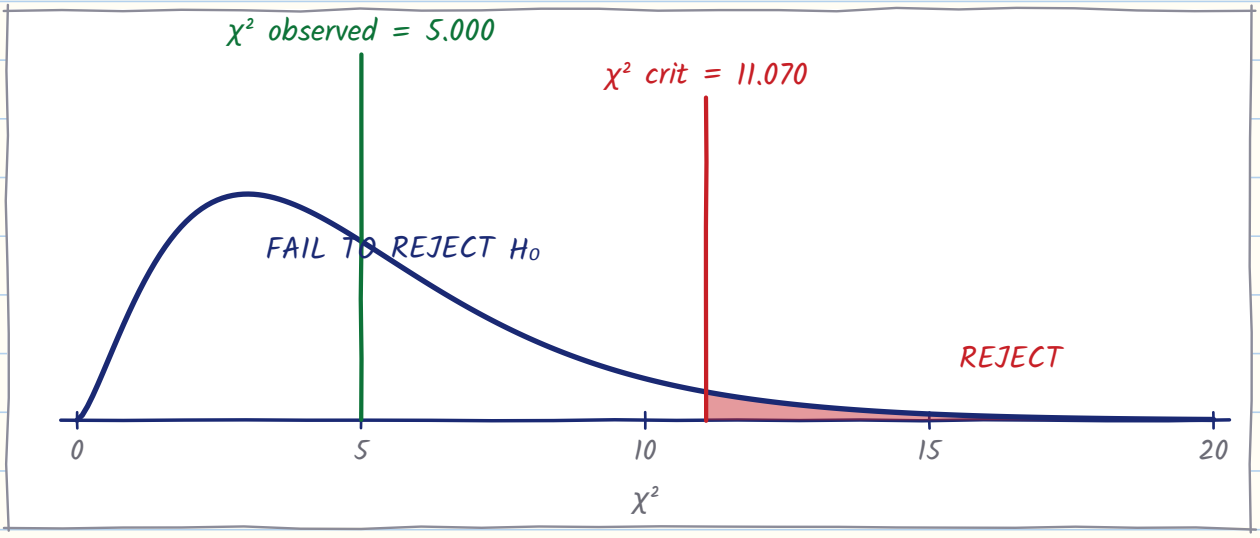


Fig 12: observed $\chi^2 = 5.0$ sits well inside the fail-to-reject zone.

B1b Goodness of fit with unequal expectations: Mendel's peas

Dihybrid cross, N = 556 plants. Theory 9 : 3 : 3 : 1 (round-yellow, round-green, wrinkled-yellow, wrinkled-green).

type	O	E = N×share	$(O-E)^2/E$
round-yellow	315	556×9/16 = 312.75	0.016
round-green	108	556×3/16 = 104.25	0.135
wrinkled-yellow	101	556×3/16 = 104.25	0.101
wrinkled-green	32	556×1/16 = 34.75	0.218
sum	556	556	$\chi^2 = 0.470$

- df = 4 − 1 = 3. χ^2 critical = 7.815. p = 0.9254 → **FAIL TO REJECT**: data fit 9:3:3:1.

```
from scipy import stats
print(stats.chisquare([25, 17, 15, 23, 24, 16]))     # equal E default
obs = [315, 108, 101, 32]
exp = [556 * w / 16 for w in (9, 3, 3, 1)]
print(stats.chisquare(obs, f_exp=exp))
# output:
# Power_divergenceResult(statistic=np.float64(5.0), pvalue=np.float64(
# 0.4158801869955079))
# Power_divergenceResult(statistic=np.float64(0.4700239808153477), pva
# lue=np.float64(0.925425895103616))
```

B2 Independence: study mode vs exam result

250 students: mode of study and whether they passed. Are mode and result related? $\alpha = 0.05$.

Observed (O) with totals

	Pass	Fail	row total
Online	45	25	70
Hybrid	60	20	80
Classroom	55	45	100
col total	160	90	250

1. H_0: mode and result are independent. H_1: they are related.

Step 2: expected counts E = row total × col total / N

	Pass	Fail
Online	70×160/250 = 44.8	70×90/250 = 25.2
Hybrid	80×160/250 = 51.2	80×90/250 = 28.8
Classroom	100×160/250 = 64.0	100×90/250 = 36.0

- Every $E \geq 5$ ✓ (smallest = 25.2).

Step 3: contributions $(O - E)^2 / E$

	Pass	Fail
Online	0.001	0.002
Hybrid	1.512	2.689
Classroom	1.266	2.250
total χ^2		7.719

4. $df = (3-1)(2-1) = \mathbf{2}$. χ^2 critical (5%) = **5.991**.
5. $\chi^2 = 7.719 > 5.991$, $p = 0.0211$ → **REJECT H_0**.
6. Cramér's $V = \sqrt{[7.719/(250\times1)]} = \mathbf{0.176}$ (small to medium).

Which cells drive it? Standardised residuals $(O-E)/\sqrt{E}$

	Pass	Fail
Online	0.03	−0.04
Hybrid	1.23	−1.64
Classroom	−1.12	1.50

CONCLUSION

Conclusion: pass rate depends on study mode ($\chi^2(2) = 7.72$, $p = 0.021$, $V = 0.18$). Classroom students fail more than expected (+1.50), hybrid students fail less (−1.64). The link is real but modest. No single residual exceeds 2; the pattern is shared.

Excel

```
Put O in B2:C4. Expected table in F2:G4.
F2  =SUM($B2:$C2)*SUM(B$2:B$4)/SUM($B$2:$C$4)    fill right and down
p   =CHISQ.TEST(B2:C4,F2:G4)                      -> 0.0211
χ²   =SUMPRODUCT((B2:C4-F2:G4)^2/F2:G4)           -> 7.7195
df  =(ROWS(B2:C4)-1)*(COLUMNS(B2:C4)-1)          -> 2
crit=CHISQ.INV.RT(0.05,2)                        -> 5.9915
p   =CHISQ.DIST.RT(χ²cell,2)                      -> same p
```

```python
import numpy as np
from scipy import stats
obs = np.array([[45, 25], [60, 20], [55, 45]])
r = stats.chi2_contingency(obs)
print('chi2 =', round(r.statistic, 3), ' df =', r.dof,
      ' p =', round(r.pvalue, 4))
print(np.round(r.expected_freq, 1))
V = np.sqrt(r.statistic / (obs.sum() * (min(obs.shape) - 1)))
print('Cramer V =', round(V, 3))
print(np.round((obs - r.expected_freq) / np.sqrt(r.expected_freq), 2))

# 2x2 with and without Yates correction
t = np.array([[12, 8], [5, 15]])
print(stats.chi2_contingency(t, correction=False).statistic)
print(stats.chi2_contingency(t, correction=True).statistic)
# output:
# chi2 = 7.719  df = 2  p = 0.0211
# [[44.8 25.2]
#  [51.2 28.8]
#  [64.  36. ]]
# Cramer V = 0.176
# [[ 0.03 -0.04]
#  [ 1.23 -1.64]
#  [-1.12  1.5 ]]
# 5.012787723785166
# 3.6828644501278776
```

12. NUMERICAL WORKING C: One-Way ANOVA

three fertilisers: every sum of squares, F, Tukey, Excel, Python

Scenario

Crop yield (kg per plot) under three fertilisers, 6 plots each. Do the mean yields differ? $\alpha = 0.05$.

plot	1	2	3	4	5	6
Fertiliser A	23	25	21	27	24	26
Fertiliser B	28	31	29	33	30	32
Fertiliser C	26	24	27	25	29	23

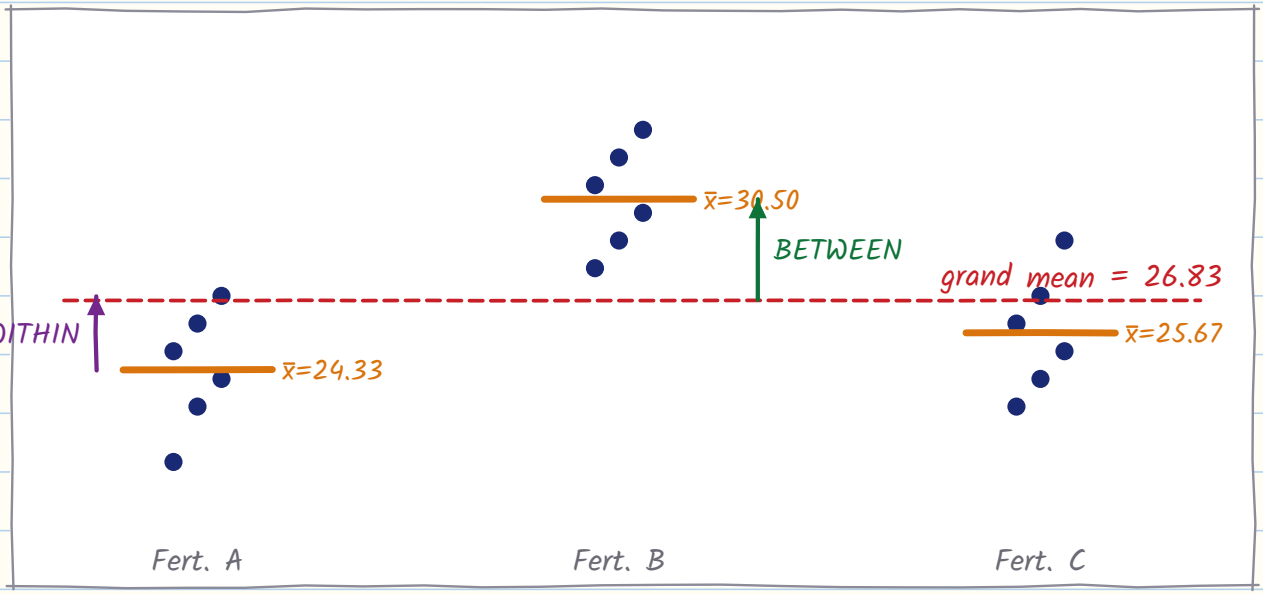


Fig 13: dots = plots, orange bars = group means, red dashes = grand mean.

Step 1: hypotheses

- H_0: $\mu_A = \mu_B = \mu_C$. H_1: at least one fertiliser mean differs.

Step 2: group summaries

group	n	Σx	mean	SS within = Σ(x−mean)²	s²
Fertiliser A	6	146	24.333	23.333	4.667
Fertiliser B	6	183	30.500	17.500	3.500
Fertiliser C	6	154	25.667	23.333	4.667

- Grand mean = (146 + 183 + 154)/18 = (483)/18 = **26.833**

Step 3: sums of squares

- SSB = $\Sigma\, n_j\,(\bar{x}_j - \text{grand})^2$ = $6[(-2.500)^2 + (3.667)^2 + (-1.167)^2]$
- = $6[6.250 + 13.444 + 1.361] = 6 \times 21.056$ = **126.333**
- SSW = 23.333 + 17.500 + 23.333 = **64.167**
- SST = SSB + SSW = 126.333 + 64.167 = **190.500** (check: $\Sigma(x - \text{grand})^2$ gives the same)

Step 4: mean squares and F

- MSB = SSB/(k−1) = 126.333/2 = **63.167**
- MSW = SSW/(N−k) = 64.167/15 = **4.2778**

$$F = \frac{63.167}{4.2778} = 14.766$$

Step 5: ANOVA table and decision

Source	SS	df	MS	F	p	F crit
Between	126.333	2	63.167	14.766	0.00029	3.682
Within	64.167	15	4.2778			
Total	190.500	17				

- F = 14.77 > F crit = 3.682 (df 2, 15) and p = 0.00029 → **REJECT H_0.**
- η^2 = SSB/SST = 126.33/190.50 = **0.663**; ω^2 = 0.605.

ANOVA CONCLUSION

Yield depends strongly on fertiliser: $F(2, 15) = 14.77$, $p < .001$, $\eta^2 = 0.66$. About 66% of yield variation is explained. NEXT: which fertilisers differ? Post-hoc.

Step 6: Tukey HSD post-hoc

- q crit (k = 3, df = 15, 5%) = 3.673. HSD = 3.673 × √(4.277816) = **3.102**

pair	mean gap	> HSD ?	Tukey p	verdict
A vs B	6.167	yes	0.0003	differ
A vs C	1.333	no	0.5189	similar
B vs C	4.833	yes	0.0028	differ

> **TIP**
>
> B beats both A and C. A and C are statistically indistinguishable. Fertiliser B is the winner.

Excel

Data: A in A2:A7, B in B2:B7, C in C2:C7 (labels in row 1).

Excel formulas

```
E2  =AVERAGE(A2:C7)                                          -> 26.833  grand mean
E3  =6*((AVERAGE(A2:A7)-E2)^2+(AVERAGE(B2:B7)-E2)^2+(AVERAGE(C2:C7)-E2)^2)
                                                            -> 126.333  SSB
E4  =DEVSQ(A2:A7)+DEVSQ(B2:B7)+DEVSQ(C2:C7)                  -> 64.167   SSW
E5  =E3/2                                                    -> 63.167  MSB
E6  =E4/15                                                   -> 4.2778  MSW
E7  =E5/E6                                                   -> 14.766  F
E8  =F.DIST.RT(E7,2,15)                                      -> 0.00029  p
E9  =F.INV.RT(0.05,2,15)                                     -> 3.682  F crit
E10 =E3/(E3+E4)                                              -> 0.663  eta squared
```

ToolPak: Anova: Single Factor

1. Data → Data Analysis → **Anova: Single Factor** → OK.
2. Input Range = A1:C7. Grouped By = **Columns**. Tick Labels in First Row. Alpha = 0.05.
3. Output Range = G1 → OK. Read 'Between Groups' and 'Within Groups' rows.

NOTE

The output also lists Count, Sum, Average and Variance for each group, then the ANOVA table with F, P-value and F crit. It does NOT run post-hoc tests.

```
import numpy as np
from scipy import stats
A = [23, 25, 21, 27, 24, 26]
B = [28, 31, 29, 33, 30, 32]
C = [26, 24, 27, 25, 29, 23]

F, p = stats.f_oneway(A, B, C)
print('F =', round(F, 3), ' p =', round(p, 5))

# assumption checks
print('Levene p =', round(stats.levene(A, B, C).pvalue, 3))
resid = np.concatenate([np.array(g) - np.mean(g) for g in (A, B, C)])
print('Shapiro p =', round(stats.shapiro(resid).pvalue, 3))

# post-hoc: Tukey HSD
t = stats.tukey_hsd(A, B, C)
print(np.round(t.pvalue, 4))

# effect size and non-parametric backup
allv = np.concatenate([A, B, C])
ssb = sum(len(g) * (np.mean(g) - allv.mean()) ** 2 for g in (A, B, C))
print('eta2 =', round(ssb / ((allv - allv.mean()) ** 2).sum(), 3))
print('Kruskal p =', round(stats.kruskal(A, B, C).pvalue, 5))
# output:
# F = 14.766  p = 0.00029
# Levene p = 0.954
# Shapiro p = 0.89
# [[1.000e+00 3.000e-04 5.189e-01]
#  [3.000e-04 1.000e+00 2.800e-03]
#  [5.189e-01 2.800e-03 1.000e+00]]
# eta2 = 0.663
# Kruskal p = 0.00416
```

13. NUMERICAL WORKING D: Two-Way ANOVA

teaching style × study time: A, B and the interaction

Scenario

Exam marks of 24 students. Factor A = teaching style (Interactive, Lecture). Factor B = weekly study time (Low, Medium, High). n = 4 students per cell. $\alpha = 0.05$.

	Low	Medium	High
Interactive	58, 66, 60, 64	70, 79, 72, 75	82, 91, 84, 87
Lecture	55, 64, 58, 63	61, 70, 64, 69	65, 75, 68, 72

Step 1: hypotheses (three pairs)

- A: H_0 no difference between teaching styles. B: H_0 no difference between study times.
- A×B: H_0 the style effect is the SAME at every study time (no interaction).

Step 2: cell, row and column means

	Low	Medium	High	row mean
Interactive	62.00	74.00	86.00	74.000
Lecture	60.00	66.00	70.00	65.333
col mean	61.00	70.00	78.00	69.667

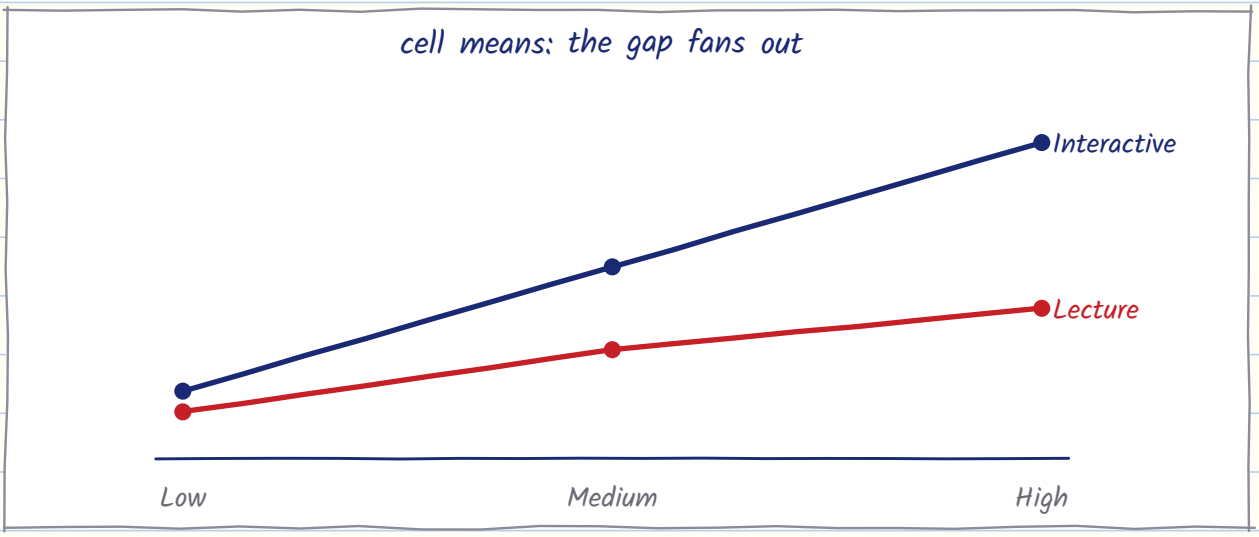


Fig 14: lines are NOT parallel, so expect an interaction.

Step 3: sums of squares

- $SSA = b \cdot n \, \Sigma(\text{row mean} - M)^2 = 3 \times 4 \times [(4.333)^2 + (-4.333)^2] =$ **450.667**
- $SSB = a \cdot n \, \Sigma(\text{col mean} - M)^2 = 2 \times 4 \times [(-8.667)^2 + (0.333)^2 + (8.333)^2] =$ **1157.333**
- Interaction residuals cell − row − col + M:

	Low	Medium	High
Interactive	−3.333	−0.333	3.667
Lecture	3.333	0.333	−3.667

- $SSAB = n \, \Sigma(\text{residual})^2 = 4 \times 49.3333 =$ **197.333**
- SSW = Σ within-cell squared deviations. Per cell:

	Low	Medium	High
Interactive	40.0	46.0	46.0
Lecture	54.0	54.0	58.0

- $SSW = 40 + 46 + 46 + 54 + 54 + 58 =$ **298.0**
- $SST = 450.7 + 1157.3 + 197.3 + 298.0 =$ **2103.3**

Step 4: ANOVA table

Source	SS	df	MS	F	p	F crit
A style	450.67	1	450.67	27.221	0.00006	4.414
B time	1157.33	2	578.67	34.953	0.000001	3.555
A×B	197.33	2	98.67	5.960	0.0103	3.555
Within	298.00	18	16.556			
Total	2103.33	23				

- df: A = 2−1 = 1, B = 3−1 = 2, A×B = 1×2 = 2, Within = 6×(4−1) = 18.
- All three effects significant. **Interaction first**: F_AB = 5.96, p = 0.0103 → REJECT.

Step 5: interpret (simple effects)

Because the interaction is significant, look at the style gap at each study time.

SE of a gap = $\sqrt{(2 \cdot MSW/n)}$ = $\sqrt{(2 \times 16.556/4)}$ = 2.877, df = 18.

study time	gap (Int − Lec)	t	p
Low	2.0	0.70	0.4958
Medium	8.0	2.78	0.0123
High	16.0	5.56	0.0000

CONCLUSION

Conclusion: teaching style and study time both matter, but style matters MORE the harder students study (gap 2 → 8 → 16 marks). At LOW study time the styles do not differ (p = 0.496). With a Bonferroni cut-off (0.05/3 = 0.0167) Medium and High remain significant.

- Partial η^2: A = 0.602, B = 0.795, A×B = 0.398.

Excel formulas (no ToolPak)

Put the 8 student rows in B2:D9: rows 2–5 Interactive, rows 6–9 Lecture. Cell means in B11:D12.

```
SST  =DEVSQ(B2:D9)                                      -> 2103.33
SSA  =12*DEVSQ(E11:E12)      (E = row means)             -> 450.67
SSB  =8*DEVSQ(B13:D13)       (row 13 = col means)        -> 1157.33
SSW  =DEVSQ(B2:B5)+DEVSQ(C2:C5)+DEVSQ(D2:D5)
     +DEVSQ(B6:B9)+DEVSQ(C6:C9)+DEVSQ(D6:D9)            -> 298.00
SSAB =SST-SSA-SSB-SSW                                   -> 197.33
F_A  =(SSA/1)/(SSW/18)       p =F.DIST.RT(F_A,1,18)
F_B  =(SSB/2)/(SSW/18)       p =F.DIST.RT(F_B,2,18)
F_AB =(SSAB/2)/(SSW/18)      p =F.DIST.RT(F_AB,2,18)
```

ToolPak: Anova: Two-Factor With Replication

1. Layout: B levels in B1:D1 (Low, Medium, High). A labels in A2 and A6. Data in B2:D9.
2. Data → Data Analysis → **Anova: Two-Factor With Replication** → OK.
3. Input Range = A1:D9. **Rows per sample = 4**. Alpha = 0.05. Output Range = G1.

Source of variation	SS	df	MS	F	P	F crit
Sample (= A)	450.67	1	450.67	27.221	5.8E-05	4.414
Columns (= B)	1157.33	2	578.67	34.953	6.3E-07	3.555
Interaction	197.33	2	98.67	5.960	0.0103	3.555
Within	298.00	18	16.556			
Total	2103.33	23				

> **TIP**
>
> In ToolPak language 'Sample' means factor A (the row blocks) and 'Columns' means factor B. Do not be confused. Also 'Without Replication' exists for n = 1 per cell (no interaction row).

Python (statsmodels, plus from scratch)

```
import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols

cells = {('Interactive', 'Low'):    [58, 66, 60, 64],
         ('Interactive', 'Medium'): [70, 79, 72, 75],
         ('Interactive', 'High'):   [82, 91, 84, 87],
```

```
        ('Lecture', 'Low'):        [55, 64, 58, 63],
        ('Lecture', 'Medium'):     [61, 70, 64, 69],
        ('Lecture', 'High'):       [65, 75, 68, 72]}
rows = [(a, b, v) for (a, b), vals in cells.items() for v in vals]
df = pd.DataFrame(rows, columns=['style', 'time', 'score'])

model = ols('score ~ C(style) * C(time)', df).fit()
print(sm.stats.anova_lm(model, typ=2).round(4))
# output:
#                      sum_sq    df        F  PR(>F)
# C(style)           450.6667   1.0  27.2215  0.0001
# C(time)           1157.3333   2.0  34.9530  0.0000
# C(style):C(time)   197.3333   2.0   5.9597  0.0103
# Residual           298.0000  18.0      NaN     NaN
```

TIP

*'C(style) * C(time)' means both main effects AND their interaction. Use '+' instead of '*' to drop the interaction. For unbalanced data prefer typ=3 with sum-to-zero contrasts.*

14. Excel Master Sheet

every function you need, plus the ToolPak menus

Function dictionary

Function	What it gives
NORM.S.DIST(z,TRUE)	area to the LEFT of z
NORM.S.INV(p)	z with area p to its left (z critical)
Z.TEST(array,x,σ)	ONE-tailed (right) p-value for a mean; x = μ_0
CONFIDENCE.NORM(α,σ,n)	margin of error $z \cdot \sigma/\sqrt{n}$
CHISQ.TEST(obs,exp)	p-value of a χ^2 test (both ranges)
CHISQ.DIST.RT(x,df)	right-tail p from χ^2
CHISQ.INV.RT(α,df)	χ^2 critical value
F.DIST.RT(F,df1,df2)	right-tail p from F
F.INV.RT(α,df1,df2)	F critical value
F.TEST(r1,r2)	p for equal variances of two ranges
DEVSQ(range)	$\Sigma(x - \text{mean})^2$ = sum of squares
SUMPRODUCT(...)	add products: handy for χ^2

WATCH OUT

Z.TEST returns the RIGHT-tail p only. For a two-tailed p use 2*MIN(Z.TEST(...),1-Z.TEST(...)).

Z-tests in Excel

```
One-sample : =(AVERAGE(A2:A37)-mu0)/(sigma/SQRT(COUNT(A2:A37)))
Two-sample : =(m1-m2)/SQRT(s1^2/n1+s2^2/n2)
One-prop   : =(x/n-p0)/SQRT(p0*(1-p0)/n)
Two-prop   : pooled  =(x1+x2)/(n1+n2)
             z      =(x1/n1-x2/n2)/SQRT(pooled*(1-pooled)*(1/n1+1/n2))
p two-tail : =2*(1-NORM.S.DIST(ABS(z),TRUE))
p right    : =1-NORM.S.DIST(z,TRUE)       p left: =NORM.S.DIST(z,TRUE)
```

Chi-square in Excel

```
GOF  : O in B2:B7, expected proportions in C2:C7
       E in D2:D7  =SUM($B$2:$B$7)*C2
       chi2 =SUMPRODUCT((B2:B7-D2:D7)^2/D2:D7)
       p    =CHISQ.TEST(B2:B7,D2:D7)    (df = k-1 built in)

Independence: O in B2:C4, E in F2:G4
       E    =SUM($B2:$C2)*SUM(B$2:B$4)/SUM($B$2:$C$4)
       p    =CHISQ.TEST(B2:C4,F2:G4)
       Cramer V =SQRT(chi2/(N*(MIN(rows,cols)-1)))
```

TIP

CHISQ.TEST gives only p. It never shows χ^2. Build χ^2 with SUMPRODUCT, or invert: =CHISQ.INV.RT(p,df).

ANOVA in Excel

```
One-way by formula:
  SSB  =n*((mean1-grand)^2+(mean2-grand)^2+...)
  SSW  =DEVSQ(group1)+DEVSQ(group2)+...
  F    =(SSB/(k-1))/(SSW/(N-k))     p =F.DIST.RT(F,k-1,N-k)
Two-way by formula: see Section 13 (DEVSQ trick on means).
```

Analysis ToolPak menu cheat

Data Analysis option	Use
Anova: Single Factor	one-way ANOVA (groups in columns)
Anova: Two-Factor With Replication	two-way with $n \geq 2$ per cell: gives interaction
Anova: Two-Factor Without Replication	$n = 1$ per cell: no interaction row
z-Test: Two Sample for Means	needs KNOWN variances as inputs
F-Test Two-Sample for Variances	compare two variances
(no χ^2 option)	use CHISQ.TEST and formulas

TIP

Enable once: File → Options → Add-ins → Manage Excel Add-ins → Go → tick Analysis ToolPak → OK. Then Data tab → Data Analysis.

15. Python Master Code

write each test yourself, then confirm with SciPy / statsmodels

Cheat list

everything in 11 lines

```
from scipy import stats
stats.norm.ppf(0.975)                          # z critical
stats.norm.sf(z)                               # right-tail p
stats.chisquare(obs, f_exp=exp)                # goodness of fit
stats.chi2_contingency(table)                  # independence
stats.f_oneway(g1, g2, g3)                     # one-way ANOVA
stats.tukey_hsd(g1, g2, g3)                    # post-hoc
stats.kruskal(g1, g2, g3)                      # rank-based ANOVA
from statsmodels.stats.proportion import proportions_ztest
proportions_ztest([x1, x2], [n1, n2])          # two-proportion z
from statsmodels.formula.api import ols    # two-way ANOVA
```

From-scratch versions (understand every line)

ztools.py

```
import numpy as np
from scipy import stats

def z_decide(z, alpha=0.05, tail='two'):
    if tail == 'two':
        p = 2 * stats.norm.sf(abs(z))
        crit = stats.norm.ppf(1 - alpha / 2)
        reject = abs(z) > crit
    elif tail == 'greater':
        p = stats.norm.sf(z)
        crit = stats.norm.ppf(1 - alpha)
        reject = z > crit
    else:                                      # 'less'
        p = stats.norm.cdf(z)
        crit = stats.norm.ppf(alpha)
        reject = z < crit
    r = lambda v: round(float(v), 4)
    return dict(z=r(z), p=r(p), crit=r(crit), reject=bool(reject))

def z_mean(xbar, mu0, sigma, n, **kw):
    return z_decide((xbar - mu0) / (sigma / np.sqrt(n)), **kw)

def z_prop1(x, n, p0, **kw):
```

```
    return z_decide((x / n - p0) / np.sqrt(p0 * (1 - p0) / n), **kw)

def z_prop2(x1, n1, x2, n2, **kw):
    pp = (x1 + x2) / (n1 + n2)                # pooled under H0
    se = np.sqrt(pp * (1 - pp) * (1 / n1 + 1 / n2))
    return z_decide((x1 / n1 - x2 / n2) / se, **kw)

def chi_gof(obs, probs, alpha=0.05):
    obs = np.asarray(obs, float)
    exp = obs.sum() * np.asarray(probs, float)
    chi2 = ((obs - exp) ** 2 / exp).sum()
    df = len(obs) - 1
    return dict(chi2=round(chi2, 4), df=df,
                p=round(stats.chi2.sf(chi2, df), 4),
                reject=bool(chi2 > stats.chi2.ppf(1 - alpha, df)))

def chi_indep(obs, alpha=0.05):
    O = np.asarray(obs, float)
    E = np.outer(O.sum(1), O.sum(0)) / O.sum()   # row x col / N
    chi2 = ((O - E) ** 2 / E).sum()
    df = (O.shape[0] - 1) * (O.shape[1] - 1)
    V = np.sqrt(chi2 / (O.sum() * (min(O.shape) - 1)))
    return dict(chi2=round(chi2, 4), df=df,
                p=round(stats.chi2.sf(chi2, df), 4), V=round(V, 3),
                reject=bool(chi2 > stats.chi2.ppf(1 - alpha, df)))

def one_way(*groups):
    g = [np.asarray(x, float) for x in groups]
    allv = np.concatenate(g)
    k, N = len(g), len(allv)
    ssb = sum(len(x) * (x.mean() - allv.mean()) ** 2 for x in g)
    ssw = sum(((x - x.mean()) ** 2).sum() for x in g)
    F = (ssb / (k - 1)) / (ssw / (N - k))
    return dict(F=round(F, 3), df=(k - 1, N - k),
                p=round(stats.f.sf(F, k - 1, N - k), 5),
                eta2=round(ssb / (ssb + ssw), 3))

def two_way(d):                    # d shape: (a, b, n), balanced
    d = np.asarray(d, float)
    a, b, n = d.shape
    M = d.mean()
    Am, Bm, C = d.mean((1, 2)), d.mean((0, 2)), d.mean(2)
    ssa = b * n * ((Am - M) ** 2).sum()
    ssb = a * n * ((Bm - M) ** 2).sum()
    ssab = n * ((C - Am[:, None] - Bm[None, :] + M) ** 2).sum()
    ssw = ((d - C[:, :, None]) ** 2).sum()
    dfw = a * b * (n - 1)
    msw = ssw / dfw
    out = {}
    for nm, ss, df in (('A', ssa, a - 1), ('B', ssb, b - 1),
                       ('AB', ssab, (a - 1) * (b - 1))):
        F = ss / df / msw
        out[nm] = (round(F, 3), round(stats.f.sf(F, df, dfw), 5))
```

```
    return out
```

Test the tools

```
print(z_mean(104.5, 100, 15, 36))
print(z_prop2(90, 600, 140, 700))
print(chi_gof([25, 17, 15, 23, 24, 16], [1 / 6] * 6))
print(chi_indep([[45, 25], [60, 20], [55, 45]]))
g = ([23, 25, 21, 27, 24, 26], [28, 31, 29, 33, 30, 32],
     [26, 24, 27, 25, 29, 23])
print(one_way(*g))
print(stats.f_oneway(*g).statistic)       # SciPy cross-check
# output:
# {'z': 1.8, 'p': 0.0719, 'crit': 1.96, 'reject': False}
# {'z': -2.3551, 'p': 0.0185, 'crit': 1.96, 'reject': True}
# {'chi2': np.float64(5.0), 'df': 5, 'p': np.float64(0.4159), 'reject'
# : False}
# {'chi2': np.float64(7.7195), 'df': 2, 'p': np.float64(0.0211), 'V':
# np.float64(0.176), 'reject': True}
# {'F': np.float64(14.766), 'df': (2, 15), 'p': np.float64(0.00029), '
# eta2': np.float64(0.663)}
# 14.766233766233771
```

TIP

The from-scratch numbers equal SciPy's. Writing them once shows what the library hides.

Test two_way on the teaching data

```
TW = np.array([
  [[58, 66, 60, 64], [70, 79, 72, 75], [82, 91, 84, 87]],
  [[55, 64, 58, 63], [61, 70, 64, 69], [65, 75, 68, 72]]])
print(two_way(TW))          # {name: (F, p)}
# output:
# {'A': (np.float64(27.221), np.float64(6e-05)), 'B': (np.float64(34.9
# 53), np.float64(0.0)), 'AB': (np.float64(5.96), np.float64(0.01032))
# }
```

NOTE

These match statsmodels anova_lm(typ=2): F_A 27.221, F_B 34.953, F_AB 5.960.

Plot the evidence

```
import matplotlib.pyplot as plt
plt.boxplot([A, B, C], labels=['A', 'B', 'C'])        # ANOVA groups
plt.show()
# interaction plot (two-way): one line per level of factor A
for name, means in {'Interactive': [62, 74, 86],
                    'Lecture': [60, 66, 70]}.items():
    plt.plot(['Low', 'Medium', 'High'], means, marker='o', label=name)
plt.legend(); plt.ylabel('mean mark'); plt.show()
```

16. Bonus: PyTorch and TensorFlow

χ^2 and F as tensor math, with ML use cases

WHY THIS EXISTS

Core PyTorch and TensorFlow ship no chi-square or ANOVA tests. But χ^2 and F are sums, means and ratios, so tensors compute them. The p-value (curve area) comes from SciPy. ML uses: data-drift checks (χ^2), A/B conversion tests (z), comparing 3+ models over seeds (ANOVA).

PyTorch

PyTorch

```
import torch
from scipy import stats

def chi2_indep_torch(obs):
    O = torch.as_tensor(obs, dtype=torch.float64)
    E = torch.outer(O.sum(1), O.sum(0)) / O.sum()   # row x col / N
    chi2 = ((O - E) ** 2 / E).sum().item()
    df = (O.shape[0] - 1) * (O.shape[1] - 1)
    return chi2, df, stats.chi2.sf(chi2, df)

def anova_f_torch(groups):
    g = [torch.as_tensor(x, dtype=torch.float64) for x in groups]
    allv = torch.cat(g)
    gm, N, k = allv.mean(), allv.numel(), len(g)
    ssb = sum(len(x) * (x.mean() - gm) ** 2 for x in g)
    ssw = sum(((x - x.mean()) ** 2).sum() for x in g)
    F = ((ssb / (k - 1)) / (ssw / (N - k))).item()
    return F, (k - 1, N - k), stats.f.sf(F, k - 1, N - k)

def z_two_prop_torch(x1, n1, x2, n2):
    x1, n1, x2, n2 = (torch.tensor(v, dtype=torch.float64)
                      for v in (x1, n1, x2, n2))
    pp = (x1 + x2) / (n1 + n2)
    se = torch.sqrt(pp * (1 - pp) * (1 / n1 + 1 / n2))
    z = ((x1 / n1 - x2 / n2) / se).item()
    return z, 2 * stats.norm.sf(abs(z))
```

TensorFlow

TensorFlow

```
import tensorflow as tf
from scipy import stats

def chi2_indep_tf(obs):
    O = tf.constant(obs, dtype=tf.float64)
    r, c = tf.reduce_sum(O, 1), tf.reduce_sum(O, 0)
    E = tf.tensordot(r, c, axes=0) / tf.reduce_sum(O)
    chi2 = float(tf.reduce_sum((O - E) ** 2 / E))
    df = (O.shape[0] - 1) * (O.shape[1] - 1)
    return chi2, df, stats.chi2.sf(chi2, df)

def anova_f_tf(groups):
    g = [tf.constant(x, dtype=tf.float64) for x in groups]
    allv = tf.concat(g, 0)
    gm = tf.reduce_mean(allv)
    N, k = int(tf.size(allv)), len(g)
    ssb = tf.add_n([tf.cast(tf.size(x), tf.float64)
                    * (tf.reduce_mean(x) - gm) ** 2 for x in g])
    ssw = tf.add_n([tf.reduce_sum((x - tf.reduce_mean(x)) ** 2)
                    for x in g])
    F = float((ssb / (k - 1)) / (ssw / (N - k)))
    return F, (k - 1, N - k), stats.f.sf(F, k - 1, N - k)
```

Check against the verified numbers

```
obs = [[45, 25], [60, 20], [55, 45]]
print(chi2_indep_torch(obs))      # or chi2_indep_tf(obs)
# expected (SciPy): chi2 = 7.719, df = 2, p = 0.0211
A = [23, 25, 21, 27, 24, 26]; B = [28, 31, 29, 33, 30, 32]
C = [26, 24, 27, 25, 29, 23]
print(anova_f_torch([A, B, C]))   # or anova_f_tf([A, B, C])
# expected: F = 14.766, df = (2, 15), p = 0.00029
print(z_two_prop_torch(90, 600, 140, 700))
# expected: z = -2.355, p = 0.0185
```

ML use cases

- **Data drift**: compare category counts in training vs live data with χ^2 independence.
- **A/B test**: click or conversion rates of two model versions → two-proportion z.
- **Model comparison**: scores of 3+ models over many seeds → one-way ANOVA, then Tukey.

- **Hyper-parameter grid**: learning-rate × batch-size effects on accuracy → two-way ANOVA.

> **WATCH OUT**
>
> Warning: scores from the same seed or fold are PAIRED, and CV folds overlap. Plain ANOVA then over-states certainty. Use repeated-measures designs or corrected resampled tests.

17. Common Mistakes and Exam Traps

what costs marks and what costs credibility

Top mistakes

1. Using z when σ is NOT known and n is small. Use t.
2. Feeding percentages instead of COUNTS into χ^2.
3. χ^2 with small expected counts ($E < 5$). Merge categories or use Fisher's exact.
4. χ^2 on paired yes/no data. Use McNemar.
5. Wrong df: GOF = $k-1$, independence = $(r-1)(c-1)$, ANOVA = $(k-1, N-k)$.
6. Using $E = N/k$ without checking H_0 really claims equal shares.
7. Reading 'significant χ^2' as 'strong link'. Report Cramér's V.
8. Claiming causation from an association test.
9. Pooling the proportions in a confidence interval (pool only in the TEST).
10. Using the sample $\hat{p}$ instead of p_0 in the one-proportion SE.
11. Running many t-tests instead of one ANOVA (inflated false alarms).
12. Reading 'significant F' as 'all groups differ'. It means at least one differs.
13. Skipping post-hoc, or post-hoc without a familywise correction.
14. Using one-way ANOVA when the SAME subjects appear in every group (repeated measures).
15. Two-way: interpreting main effects while a big interaction is present.
16. Two-way: unequal cell sizes without choosing Type II or III sums of squares.
17. Looking at BOTH tails of χ^2 or F. Only the right tail is used.
18. Reporting F without both df, or χ^2 without df.

Exam tricks

- Variance given? Square the SD or root the variance. Check the units first.
- ANOVA table with a missing cell? Use SST = SSB + SSW and MS = SS/df to fill it.
- F is always ≥ 0. Negative F means arithmetic error.
- 2×2 χ^2 and two-proportion z agree: $z^2 = \chi^2$ (without Yates).

- Two groups in ANOVA? $F = t^2$, same p-value as the pooled t-test.

FINAL CHECK

30-second self-check: (1) data type? (2) right test and tail? (3) df right? (4) assumptions? (5) conclusion in plain words with effect size?

18. Formula Cheat Sheet and Tables

one-page recall and the critical-value tables

Test	Statistic	df
One-sample z	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	—
Two-sample z	$z = (\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$	—
One-prop z	$z = (\hat{p} - p_0)/\sqrt{[p_0(1-p_0)/n]}$	—
Two-prop z	$z = (\hat{p}_1 - \hat{p}_2)/\sqrt{[\hat{p}(1-\hat{p})(1/n_1+1/n_2)]}$	—
χ^2 GOF	$\chi^2 = \Sigma\ (O - E)^2/E$, $E = N \times$ share	$k - 1$
χ^2 independence	$\chi^2 = \Sigma\ (O - E)^2/E$, $E = R \cdot C/N$	$(r-1)(c-1)$
One-way ANOVA	$F = [SSB/(k-1)]\ /\ [SSW/(N-k)]$	$(k-1,\ N-k)$
Two-way ANOVA	$F_{eff} = MS_{eff}\ /\ MS_W$ (A, B, A×B)	see below
Effect sizes	$\eta^2 = SSB/SST$, $V = \sqrt{[\chi^2/(N \cdot \min(r-1,c-1))]}$	—

- Two-way df: A = a−1, B = b−1, A×B = (a−1)(b−1), Within = ab(n−1), Total = abn−1.

Excel and SciPy quick map

Test	Excel	SciPy / statsmodels
z (p-value)	2*(1-NORM.S.DIST(ABS(z),TRUE))	stats.norm.sf
2-prop z	build z, then NORM.S.DIST	proportions_ztest
χ^2 GOF	CHISQ.TEST(O,E)	stats.chisquare
χ^2 independence	CHISQ.TEST(O,E)	stats.chi2_contingency
One-way ANOVA	ToolPak Single Factor	stats.f_oneway
Two-way ANOVA	ToolPak Two-Factor w/ Repl.	ols + anova_lm

χ^2 critical values (right tail)

df	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1	2.706	3.841	6.635
2	4.605	5.991	9.210
3	6.251	7.815	11.345

4	7.779	9.488	13.277
5	9.236	11.070	15.086
6	10.645	12.592	16.812
7	12.017	14.067	18.475
8	13.362	15.507	20.090
9	14.684	16.919	21.666
10	15.987	18.307	23.209
12	18.549	21.026	26.217
15	22.307	24.996	30.578
20	28.412	31.410	37.566
25	34.382	37.652	44.314
30	40.256	43.773	50.892
40	51.805	55.758	63.691
50	63.167	67.505	76.154
100	118.498	124.342	135.807

F critical values, $\alpha = 0.05$ (right tail)

$df_2 \setminus df_1$	1	2	3	4	5	6	8	10	12
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.74	4.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.35	3.28
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.98	2.91
12	4.75	3.89	3.49	3.26	3.11	3.00	2.85	2.75	2.69
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.54	2.48
18	4.41	3.55	3.16	2.93	2.77	2.66	2.51	2.41	2.34
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.35	2.28
25	4.24	3.39	2.99	2.76	2.60	2.49	2.34	2.24	2.16
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.16	2.09
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.08	2.00
60	4.00	3.15	2.76	2.53	2.37	2.25	2.10	1.99	1.92
120	3.92	3.07	2.68	2.45	2.29	2.18	2.02	1.91	1.83

RULE

Decision rule for χ^2 and F: statistic > critical value → reject H_0. Equivalent to $p < \alpha$.

19. EXERCISE BANK

10 problems covering z, chi-square, one-way and two-way ANOVA

HOW TO ATTEMPT

For each: (1) pick the test, (2) state H_0/H_1, (3) compute the statistic and df, (4) compare with the critical value or p, (5) decide in words. $\alpha = 0.05$. Key is on the last page.

Q1 Bottling line (σ known)

A line has known $\sigma = 2.5$ ml. A sample of $n = 49$ bottles has mean 499.2 ml. The label says 500 ml. Is the true mean LESS than 500 ml? (one-tailed, left)

Q2 Two exam boards (σ known)

Board 1 ($\sigma = 6$): $n = 60$, mean 78. Board 2 ($\sigma = 8$): $n = 80$, mean 75.5. Do the means differ? (two-tailed)

Q3 Drug vs placebo

78 of 120 patients on the drug recovered; 60 of 110 on placebo recovered. Is the recovery rate different? (two-tailed)

Q4 Defect rate claim

A factory claims a 5% defect rate. In a sample of 400 items, 28 are defective. Is the true rate GREATER than 5%? (one-tailed, right)

Q5 Customer arrivals by weekday

Observed customers Mon to Fri: 48, 62, 55, 40, 45 ($N = 250$). Is arrival uniform across the five weekdays?

Q6 Vaccination and infection (2×2)

Vaccinated: 30 infected, 70 not infected. Unvaccinated: 50 infected, 50 not infected. Is infection related to vaccination? (use plain Pearson χ^2, no Yates correction)

Q7 Region and preferred plan (2×3)

	Plan A	Plan B	Plan C
Urban	20	30	50
Rural	40	30	30

Is plan preference related to region?

Q8 Three diets (weight loss, kg)

Do the mean weight losses differ? n = 5 each.

Diet 1: 4.2, 5.1, 3.8, 4.9, 5.5

Diet 2: 6.1, 7.4, 6.8, 7.9, 6.5

Diet 3: 5.0, 5.8, 4.6, 5.2, 6.1

Q9 Four teaching methods (summary stats)

method	M1	M2	M3	M4
mean	50	55	53	58
SD	6	5	7	6
n	10	10	10	10

Do the four method means differ?

Q10 Fertiliser × watering (two-way, 3 plots per cell)

	Low water	High water
Fertiliser F1	14, 16, 15	20, 22, 21
Fertiliser F2	18, 17, 19	19, 21, 20

Test the main effect of fertiliser, the main effect of watering, and their interaction.

> **TIP**
>
> Stuck choosing? Re-read Section 9: first decide 'measurement or count?', then count the groups and factors.

20. Answer Key

check your work: test type, statistic, decision

Verify only AFTER you have attempted every problem.

Q	Test type	statistic	Decision
Q1	One-sample z (left-tailed)	$z = -2.240$	Reject H_0
Q2	Two-sample z, means	$z = 2.113$	Reject H_0
Q3	Two-proportion z	$z = 1.617$	Fail to Reject H_0
Q4	One-proportion z (right-tailed)	$z = 1.835$	Reject H_0
Q5	Chi-square goodness of fit	$\chi^2 = 5.960$, df 4	Fail to Reject H_0
Q6	Chi-square independence (2×2)	$\chi^2 = 8.333$, df 1	Reject H_0
Q7	Chi-square independence (2×3)	$\chi^2 = 11.667$, df 2	Reject H_0
Q8	One-way ANOVA (raw data)	$F = 14.726$, df 2, 12	Reject H_0
Q9	One-way ANOVA (summary stats)	$F = 3.105$, df 3, 36	Reject H_0

Q10 Two-way ANOVA (df = 1, 8 for every effect)

Effect	F	p	Decision
A Fertiliser	3.00	0.1215	Fail to Reject H_0
B Watering	48.00	0.0001	Reject H_0
A×B	12.00	0.0085	Reject H_0

Verification notes

- Critical values: Q1 −1.645, Q4 +1.645 (one-tailed); Q2, Q3 ±1.960.
- χ^2 critical: Q5 9.488 (df 4), Q6 3.841 (df 1), Q7 5.991 (df 2).
- Q6 with Yates correction: $\chi^2 = 7.521$ (still rejects). Also z^2 of the two-proportion test = χ^2.
- F critical: Q8 3.885 (df 2, 12), Q9 2.866 (df 3, 36), Q10 5.318 (df 1, 8).
- Q9 is borderline: $p = 0.039$. Q3 fails (two-tailed $p = 0.106$) despite a 10-point gap.

TIP

Got a different value? Check: σ vs s, counts vs percentages, df, SS vs MS, and whether you divided the right row/column totals for E.

~ end of notes ~

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